Risk-Aware Market Clearing in Electricity Markets

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1 Introduction

~~In this work, we study the design of a centralized market under uncertainty. Our focus is on two-stage markets, where a decision in the first stage is made based on an uncertain second stage. Electricity markets serve as a canonical example of such a two-stage market.~~The design of electricity markets serves as a canonical example of a two-stage centralized market under uncertainty, where a proactive decision must be taken in the first stage, in anticipation of future reactive decisions after resolution of the inherent uncertainty in the second stage. Although structurally different, both markets in the United States and Europe operate in two stages: the Day-Ahead (DA) and the Real-Time (RT) market. In the DA market, market agents (electricity buyers and sellers) submit bids and offers, and an Independent System Operator (ISO) clears the market, publishing allocations (i.e., a tentative commitment and dispatch schedule) and day-ahead prices. The bids and offers submitted by agents in the DA market are based on uncertain forecasts of load and generation, with uncertainty arising from, for example, the weather, which translates into uncertain forecasts of load and generation of renewable energy sources. Closer to the time of delivery, the RT market clears any imbalances that arise from forecast errors or unexpected outages. This makes advance scheduling inseparable from the cost of subsequent balancing actions. Two-stage stochastic market-clearing models address this dependence by coordinating advance decisions with scenario-dependent real-time adjustments. Pritchard et al. [17] developed an energy-only settlement framework for unpredictable participants and established revenue adequacy in expectation. Building on this model, Zavala et al. [26] analyzed price consistency and showed how recourse penalties control differences between day-ahead and expected real-time prices. In this work, we extend this stochastic-clearing framework to incorporate risk.

[Amy: *discuss what a stochastic competitive equilibrium is next, and cite past work that has looked at SCE in electricity markets.*]

A natural way to study such uncertain two-stage markets is through the lens of *stochastic competitive equilibrium* (SCE), in which agents jointly optimise their first- and second-stage decisions, taking prices as given, while supply equals demand in the first and second stages of the market in every possible second-stage scenario. This is the analogue of classical competitive equilibrium [18] under uncertainty, specialized to two stages. In electricity markets, SCE has been previously studied in relation to day-ahead and real-time settlement structures. ~~Pritchard et al designed a single-settlement market using a two-stage stochastic program that clears efficiently despite unpredictable generation, and Wong and Fuller formulated a stochastic ISO pricing problem for energy and reserves.~~ [Amy: *you seem to have missed this citation (find the real publication outlet): https://optimization-online.org/wp-content/uploads/2016/06/5484.pdf*][Hrithik: *cited zakeri2019 below*] Prior works[26][25] assume that agents are risk-neutral. In the risk-neutral case, agents evaluate these two-stage decisions by optimising for their expected payoffs. [Amy: *tie this idea of risk-neutrality back to the earlier def'n of SCE, which said markets optimize in expectation.*][Hrithik: *In the definition of SCE, I mentioned that the agents jointly optimize two-stage decision, but haven't mentioned how (in expectation or risk-aware). Should we define SCE as optimizing in expectation?*]

What makes the design of electricity markets particularly challenging, and distinct from typical two-stage stochastic optimization problems like the newsvendor problem [5], and even hydrothermal electricity systems [16], is that electricity cannot easily be stored at scale, so inventory cannot be used to bridge the gap between demand and supply from the first stage to the second. [Ben: *You might want to point to the California "Duck" problem – i.e. even in California where they have invested in 17GW of storage they still can't store nearly enough to handle the grid demand spike as the sun sets. see: https://www.energy.ca.gov/news/2025-11/californias-battery-storage-fleet-continues-record-growth-strengthening-grid and https://www.eia.gov/todayinenergy/detail.php?id=56880*] As a

result, if real-time supply or demand does not match the corresponding day-ahead commitment, the difference has to be corrected by ramping generation up or down ~~, or by curtailing demand~~ [Amy: *how can you curtail demand? is that a variable in our model? or can you just ramp up or ramp down supply to meet whatever the demand is ultimately*] [Hrithik: *In our model (and also Zavala's) we allow recourse for demand side. For implementation, since demand is mostly treated as inelastic we impose very high recourse penalty. But the model is more general.*]in every scenario. This is why it is especially important to consider the risk preferences of agents, and not just expected welfare or cost in this market.

[Amy: *is it at every node? or in every possible second-stage scenario?*] [Hrithik: *Its both at every node and every scenario. This extends from Zavala s model. Since we don't have nodal balance enforced in this paper, I have mentioned only "in every possible scenario" above.*]

These challenges are becoming increasingly relevant as variable renewable resources take on a growing share of generation. Wind and solar generators together supplied roughly 17% of global electricity generation in 2025, up from about 4% a decade earlier, and renewables overall are projected to reach 43% of global generation by 2030 [12]. Since renewable sources are dependent on weather, this growth has measurably fattened the tails of real-time forecast errors [11]. A market design that clears only in expectation, by construction, e.g., the aforementioned stochastic market-clearing models, does not account for this tail. Philpott et al. [16] studied hydro-thermal electricity systems, extending the risk-neutral SCE framework to allow each agent to evaluate uncertain outcomes using a coherent risk measure, e.g., conditional value at risk (CVaR), [Amy: *or should this say mean-CVaR?*] rather than plain expectation. Their work shows that a risk-adjusted social optimum still coincides with a competitive equilibrium, provided agents can trade a sufficiently rich set of state-contingent instruments, i.e., Arrow-Debreu securities, to hedge against inflow uncertainty. ~~Gérard et al. [9] develop the underlying theory more abstractly, formalizing risk-averse comptitive equilibriu~~ [Amy: *i'm not sure this is accurate. i think the Philpott paper already did this. indeed, you wrote, "provided agents can trade a sufficiently rich set of state-contingent instruments to hedge against inflow uncertainty," which amounts to assuming a complete market. also, the Gerard work is very concrete!!, not abstract!! how about:*] ~~Gérard et al. [9] study equilibrium computation in an incomplete variant of this market.~~ [Amy: *but tbh, this paper doesn't actually seem worth mentioning to me...*] [Hrithik: *I agree, we should skip this paper in the introduction.*]

[Hrithik: *I am presenting our work as an extension of Zavala rather than building on Philpott.*] We develop a risk-aware extension of the stochastic market-clearing formulation of Zavala et al. [26]. Our model retains the coupling between day-ahead commitments and scenario-dependent real-time adjustments, including the recourse penalties, while replacing expected welfare of the market with a mean–CVaR coherent risk objective. We ask whether the resulting centralized allocation can be supported by electricity prices and financial transfers when consumers and producers independently optimize their risk-adjusted payoffs. Under complete Arrow–Debreu trading, we establish the correspondence between centralized clearing and competitive equilibrium for this model and give explicit constructions of supporting prices and insurance transfers. We also characterize the effective planner risk criterion in a market with agents with heterogeneous risk aversion, and construct insurance transfers that support the corresponding allocation. Our analysis makes the implications of risk-trading presented in Philpott et al. [16] and Gérard et al. [9] explicit for the risk-aware stochastic market-clearing model studied in this paper.

Main Contributions. We develop a risk-aware extension of the stochastic market-clearing formulation of Zavala et al. [25], incorporating mean–CVaR preferences into a two-stage market with day-ahead commitments and scenario-dependent real-time adjustment costs. Building on the complete-market risk-trading framework of Philpott et al. [16] and Gérard et al. [9], we make

four contributions: (i) a unified formulation connecting centralized risk-aware clearing with the individual optimization problems of consumers and producers; (ii) constructive first and second welfare theorems for this settlement model, including explicit dual-based formulas for day-ahead, real-time, and Arrow–Debreu security prices, together with supporting insurance portfolios; (iii) a characterization of the effective planner confidence level when participants have heterogeneous confidence levels and a common mean–CVaR weight, with an equilibrium construction that assigns residual aggregate risk to the participants with the lowest confidence level; and (iv) an evaluation on synthetic markets and scenarios derived from IM3 GO WEST[2], quantifying the effects of risk-aware clearing on dispatch, expected welfare, and tail welfare. Our emphasis is on the explicit connection between physical allocations, settlements, and individual risk exposure in the day-ahead and real-time clearing model.

2 Electricity Market Design (US vs EU)

Electricity market design differs considerably between the United States and Europe, in the degree of centralization, the treatment of network constraints, and the division of responsibilities between the market operators and the system operators.

[Amy: *what does "security-constrained" mean?*] [Hrithik: *It refers to grid security - the commitment schedule must also remain feasible under the loss of any single transmission line or generator(called the "N-1" contingency).*]

In the United States, ISOs are responsible for both market clearing and system operations. They centrally run both day-ahead and real-time markets by solving two interconnected optimisation problems. First, they solve a unit commitment problem, referred to as security-constrained unit commitment (SCUC) to determine which generators should be online in advance, i.e., who should commit the day before to meeting forecasted load, [Amy: *please feel free to edit, but i think for clarity the wording should mention: 1. commitment, and 2. day ahead*] while accounting for nonconvex operational constraints such as ramp-up rates, minimum up and down times, startup costs, and planned outages. This problem is typically posed and solved as a mixed integer program. Second, they solve the security-constrained economic dispatch (SCED) problem to determine how much each committed generator should produce to meet the realized demand (i.e., in real time) [Amy: *again, i think we have to link SCED to the real-time market*] at minimum cost. This dispatch problem is fundamentally a market-clearing problem. It matches supply and demand, subject to network feasibility. Generators submit supply offers, typically in the form of stepwise price-quantity bids, and demand is represented through ISO demand forecasts provided by various load-serving entities (LSEs). In most US formulations, demand is treated as inelastic, meaning the load is treated as "must serve," rather than as responsive to market prices. [MaxBoon: *citation needed*][Hrithik: *Not sure what a good citation is for this.*] As a result, the SCUC day-ahead ISO objective is typically expressed as cost minimization subject to market-clearing and transmission constraints. Then, the SCED in the real-time market adjusts to the SCUC solution and the realized system conditions in short intervals, often every 5 to 15 minutes.

The US market-clearing model is also distinguished by its use of nodal pricing. A node is a location in the transmission system~~, typically a bus or an aggregate injection or withdrawal point,~~ at which power can be injected by generators, withdrawn by loads, or transferred through transmission lines. The transmission network is represented explicitly through nodes and lines, and power balance is enforced at each of these nodes. The real-time market adjusts ~~for any deviations~~the system's dispatch and enforces power balance at all individual nodes and computes corresponding per-node Locational Marginal Prices (LMPs).

[Amy: *is power balance required?*] [Hrithik: *In the current implementation looks like equality is enforced. However, recent research works have been suggesting chance constraint and I have discussed*

this in related work (paragraph 4).] [Amy: *future work? or simply related? i.e., already done?*][Hrithik: *Implications to prices already discussed by Dvorkin et al. i.e. already done*]

The European market design is organized differently; it is zonal rather than nodal. The European day-ahead market is a centralized auction: market participants submit buy and sell orders before gate closure, after which a common algorithm [7] matches orders, sets zonal prices, and schedules cross-zonal exchanges, subject to cross-zonal transmission constraints. After this auction clears, market participants continue to trade electricity commitments in a decentralized intraday market up until delivery time, rebalancing their positions as new information arrives.

Each of these market designs has its own merits. The U.S. approach yields strong locational signals and integrates network feasibility directly into market clearing through nodal balance and transmission constraints. The European approach, by contrast, is naturally framed as a welfare-maximizing exchange and supports broad cross-border participation through a simpler zonal trading architecture. The main advantage of a welfare-maximization objective in a market is that it gives buyers and sellers the opportunity respond to market prices. The "must-serve" nature of US electricity markets means that load cannot respond to high prices during scarcity events.

Our contribution in this paper is orthogonal to the market design question. We study how risk preferences should be priced in a two-stage electricity market. To isolate that question, our formal model treats electricity as a single divisible good, with no explicit network, nodes, or zones. ~~We therefore take no position on whether nodal or zonal pricing is preferable. We believe that is a separate design question our results do not address.~~ Extending the risk-aware clearing framework developed here to a network-constrained setting, nodal or zonal, is a natural direction for future work.

3 Related Work

The classic algorithm for clearing an electricity market in the United States is based on a deterministic version of economic dispatch whose dual variables yield locational marginal prices (LMPs)~~supporting~~, which, under idealized conditions, support a competitive equilibrium. The growing penetration of renewables has prompted a variety of generalizations of this design that internalize uncertainty within the clearing problem. Our work belongs to this line of research, but departs from risk-neutral stochastic clearing by applying conditional value-at-risk (CVaR) [19] to the distribution of welfare losses in a two-stage day-ahead and real-time market.

Early stochastic market-clearing models include Wong and Fuller [24] , who proposed a two-stage stochastic linear program for joint energy and reserve dispatch under contingency uncertainty. Pritchard et al. [17] introduced a stochastic clearing approach with a single-settlement energy-only market. Morales et al. [15] show that clearing the day-ahead market at the expected value of stochastic production is inefficient; moreover, while a fully stochastic dispatch maximizes market efficiency, it does not guarantee cost recovery and can leave some producers with losses. Zavala et al. [26] extended this work by focusing on pricing consistency; deterministic clearing can create arbitrary distortions between day-ahead prices and expected real-time prices, while a stochastic formulation bounds these distortions and can eliminate them under appropriate penalty structures. Inspired by the aforementioned work, we formulate market clearing as a two-stage stochastic program with the first-stage as day-ahead commitments and the second-stage as real-time recourse. This setup allows the planner to preserve the two-stage stochastic structure, while explicitly prioritizing adverse welfare outcomes in the tail of the scenario distribution. [Amy: *after describing all of this, we need to isolate how our model differs from previous work!!*]

Recent pricing work also studies uncertainty in multi-period dispatch. Guo et al. [10] analyze pricing multi-interval dispatch under uncertainty, focusing on dispatch-following incentives for generators in settings with intertemporal operational constraints. Werner et al. [23] also study

pricing uncertainty in stochastic multi-stage electricity markets, developing prices for uncertainty across a multi-stage setting rather than only for deterministic energy quantities. Extending our framework to multi-stage settings, by integrating the explicit price identifications and heterogeneity results developed here with multi-stage pricing structures is a natural direction for future work. [Amy: *my take is that Philpott is also a multi-stage problem. maybe this isn't a natural direction for future work for us, after all? not sure?*]

Another branch of relevant literature concerns risk-sensitive dispatch models. Dvorkin et al. [8] and Mieth et al. [14] developed a stochastic market in which uncertainty is internalized through chance constraints on generator and network limits, with energy and reserve prices recovered from the dual of the resulting program. They introduce Arrow–Debreu securities into their chance-constrained stochastic electricity market to enable risk trading. Madavan et al. [13] propose a risk-sensitive SCED model in which resilience to line failures is represented using CVaR. ~~The main difference between their setting and ours is that theirs is security-constrained dispatch with contingency risk,~~[Amy: ~~*what is "contingency risk"? is this an AD-security thing? unfortunately, those ideas hav*~~ ~~whereas ours is a two-stage stochastic market-clearing model in which the risk measure is applied to social welfare loss.~~Their risk measure is used to hedge operational violations following network contingencies, whereas ours is applied directly to aggregate welfare losses across renewable and load scenarios. Tapia et al. [22] propose market clearing for extreme events using large-deviation-theory chance constraints and weighted chance constraints. Like us, they treat extreme events as a distinct objective of market design rather than as ordinary low-probability scenarios. Their approach introduces a new, separately-procured extreme reserve product, and uses large deviation theory to locate the dominating point of the tail. Our approach differs in that we use risk measures to implicitly tilt the social welfare objective to prioritize tail events and introduce Arrow–Debreu insurance securities, which are purely financial instruments.

The theoretical foundation for risk trading goes back to Arrow's theory of securities [4] and optimal risk-bearing. Arrow showed that state-contingent securities can support efficient allocation of risk under uncertainty, establishing the conceptual basis for completing markets through claims contingent on realized states. Gerard et al. [9] study risk-averse competitive partial equilibrium in stochastic electricity markets where agents are endowed with coherent risk measures. They distinguish risk-neutral equilibrium, risk-averse equilibrium, and risk-trading equilibrium with Arrow–Debreu securities. Their main message is directly relevant to our setting: with coherent risk measures, a complete risk market can support an efficient risk-adjusted social optimum, but when the risk market is incomplete, risk-averse competitive equilibria need not be unique and may fail to coincide with the social planner's solution. Our work specializes this abstract framework to a two-stage day-ahead and real-time electricity market with multiple agents and mean–CVaR preferences, deriving explicit price formulas and a constructive transfer rule. We further develop the heterogeneous-risk-preference case in detail, showing that more risk-averse agents are endogenously fully insured, while less risk-averse agents bear all residual aggregate risk. [Amy: *can we make this sound more exciting?! surprising somehow? less incremental, etc.*] [Amy: *this contribution of ours is fine here, but we should make sure that every contribution we describe in this Related Work section is first/also explained in the intro/contributions section.*][Hrithik: *I have explained our contributions in the introduction.*][Amy: *where? which point? 1,2,3, or 4?*]

4 Risk Measures

A risk measure assigns a real number to a random variable representing loss, summarizing in a single value the riskiness of holding that random loss. In our setting, the random loss in question is the negative second-stage welfare, which depends on the realized scenario. Different choices of

risk measure correspond to different attitudes toward uncertainty, and the structural properties of the risk measure determine whether it can be decentralized through prices.

This section reviews the framework of coherent risk measures [6], introduces Conditional Value-at-Risk (CVaR) as a coherent risk measure, and motivates the convex combination of expectation and CVaR that we adopt in Section 8.1.

4.1 Coherent Risk Measures

Let Ω be a finite scenario set with probability measure P assigning $\rho_\omega > 0$ to each $\omega \in \Omega$.[Ben: *added* $\in \Omega$] Let $\mathcal{X}$ denote the space of real-valued random variables on Ω, interpreted as losses (larger values are worse).

Definition 1 (Coherent Risk Measure, 6). *A function $\rho : \mathcal{X} \to \mathbb{R}$ is a* coherent risk measure *if it satisfies the following four axioms for all $X, Y \in \mathcal{X}$ and $a \in \mathbb{R}$, $\alpha \geq 0$:*

- **(M)** Monotonicity*: if $X \leq Y$ pointwise, then $\rho(X) \leq \rho(Y)$.*
- **(T)** Translation equivariance*: $\rho(X + a) = \rho(X) + a$.*
- **(P)** Positive homogeneity*: $\rho(\alpha X) = \alpha\, \rho(X)$.*
- **(S)** Subadditivity*: $\rho(X + Y) \leq \rho(X) + \rho(Y)$.*

Coherent risk measures admit a fundamental dual characterization.

Lemma 1 (Dual Representation, 6). *A function $\rho : \mathcal{X} \to \mathbb{R}$ is a coherent risk measure if and only if there exists a closed convex set $\mathcal{Q}$ of probability measures on Ω, each absolutely continuous with respect to P, such that*

$$\rho(X) \;=\; \sup_{Q \in \mathcal{Q}} \mathop{\mathbb{E}}_{Q}[X], \qquad \forall X \in \mathcal{X}. \tag{1}$$

The set $\mathcal{Q}$ is called the risk set *of ρ, and any $Q \in \mathcal{Q}$ attaining the supremum at X is called a* worst-case measure *for X.*

A standard useful property is that convex combinations of coherent risk measures are coherent [1]. We include a short proof below since our mean-CVaR welfare objective is exactly such a combination.

Lemma 2 (Convex combinations preserve coherence). *Let ρ_1, ρ_2 be coherent risk measures with risk sets $\mathcal{Q}_1, \mathcal{Q}_2$. For any $\alpha \in [0, 1]$, the function $\rho := \alpha\, \rho_1 + (1 - \alpha)\, \rho_2$ is a coherent risk measure with risk set*

$$\mathcal{Q} \;=\; \{\alpha Q_1 + (1 - \alpha) Q_2 : Q_1 \in \mathcal{Q}_1,\ Q_2 \in \mathcal{Q}_2\}.$$

Proof. Each axiom (M), (T), (P), (S) is preserved under convex combinations: (M), (T), (P) are linear in ρ, and (S) is convex (a convex combination of sub-additive inequalities is sub-additive).

For the risk set, we proceed in three steps. First, since the two suprema are over independent choices, the sum of the suprema equals the supremum of the sum:

$$\alpha \sup_{Q_1 \in \mathcal{Q}_1} \mathop{\mathbb{E}}_{Q_1}[X] + (1 - \alpha) \sup_{Q_2 \in \mathcal{Q}_2} \mathop{\mathbb{E}}_{Q_2}[X] = \sup_{Q_1 \in \mathcal{Q}_1,\, Q_2 \in \mathcal{Q}_2} \big(\alpha \mathop{\mathbb{E}}_{Q_1}[X] + (1 - \alpha) \mathop{\mathbb{E}}_{Q_2}[X]\big).$$

Second, since expectation is linear in the probability measure, $\alpha\, \mathbb{E}_{Q_1}[X] + (1 - \alpha)\, \mathbb{E}_{Q_2}[X] = \mathbb{E}_{\alpha Q_1 + (1-\alpha) Q_2}[X]$, where $\alpha Q_1 + (1 - \alpha) Q_2$ is again a probability measure. Third, reparametrising $Q := \alpha Q_1 + (1 - \alpha) Q_2$ as (Q_1, Q_2) ranges over $\mathcal{Q}_1 \times \mathcal{Q}_2$, the supremum ranges exactly over $\mathcal{Q} = \{\alpha Q_1 + (1 - \alpha) Q_2 : Q_1 \in \mathcal{Q}_1,\ Q_2 \in \mathcal{Q}_2\}$, yielding $\sup_{Q \in \mathcal{Q}} \mathbb{E}_Q[X]$ as claimed. □

4.2 Conditional Value-at-Risk

We now specialize the framework of 4.1 to Conditional Value-at-Risk (CVaR), which is the specific coherent risk measure we use throughout the paper. We present its definition and key optimization properties for general loss distributions, following Rockafellar and Uryasev [19]; further, the general-distribution treatment of Rockafellar and Uryasev [20] is essential here because our market model has finitely many scenarios, so the loss distribution is discrete.

Let $f(x, y)$ be loss associated with $x \in X \subseteq \mathbb{R}^n$, $y \in \mathbb{R}^m$. For example, x is the social welfare, y is the uncertainties involved, for example, the market parameters. Let $p(y)$ denote the probability density of y.

Let's define the probability of $f(x, y)$ not exceeding α as,

$$\Psi(x, \alpha) = \int_{\{f(x,y) \leq \alpha\}} p(y)\, dy = \mathbb{P}(f(x, y) \leq \alpha)$$

[Hrithik: *All the proof below is from Rockefellar 2000. There is similarly a proof in their follow-up work that shows that the linear program works for any general loss function which includes a discrete probability measure which is relevant to our model.*]

Assumption: The probability distribution is such that no jumps occur, i.e., $\Psi(x, \alpha)$ is everywhere continuous w.r.t. α. Then,

$$\beta\text{-VaR is defined as} : \alpha_\beta(x) = \min\{\alpha \in \mathbb{R} : \Psi(x, \alpha) \geq \beta\}$$

$$\beta\text{-CVaR is defined as} : \Phi_\beta(x) = \frac{1}{1-\beta} \int_{\{f(x,y) \geq \alpha_\beta(x)\}} f(x, y)\rho(y)\, dy$$

~~The important work by [19] is the characterization of $\Phi_\beta(x)$ and $\alpha_\beta(x)$ in terms of a new function,~~Importantly, Rockafellar and Uryasev [19] provide a characterization of $\alpha_\beta(x)$ and $\Phi_\beta(x)$ in terms of a new function,

$$\Rightarrow F_\beta(x, \alpha) = \alpha + \frac{1}{1-\beta} \int_{\mathbb{R}^m} [f(x, y) - \alpha]^+ \rho(y)\, dy$$

Theorem 1 (Theorem 01 (Rockafellar and Uryasev [19])). *As a function of α, $F_\beta(x, \alpha)$ is convex & continuously differentiable.*
β-CVaR of $f(x, y)$ is $\Phi_\beta(x) = \min_{\alpha \in \mathbb{R}} F_\beta(x, \alpha)$, and β-VaR is $\alpha_\beta(x) = \inf \arg\min_{\alpha \in \mathbb{R}} F_\beta(x, \alpha)$

[Ben: *This section appears to nest the def/proof of lemma 3, assumption 2.1, 2.2, and prop 2.1 inside the proof of theorem 1. You don't want nesting. Reorganize so the parts are presented first (defer proofs to an appendix if they break the flow), and then present the theorem and its proof after. you don't want to nest needed parts or proofs inside the main proof – they should be presented previously and distinctly.*]

Proof.

$$F_\beta(x, \alpha) = \alpha + \frac{1}{1-\beta} \int_{\mathbb{R}^m} [f(x, y) - \alpha]^+ \rho(y)\, dy$$

We assume that $\Psi(x, \alpha)$ is continuous w.r.t. α, i.e., regardless of the choice of x, the set of y with $f(x, y) = \alpha$ has probability 0, i.e.,

$$\int_{\{f(x,y)=\alpha\}} \rho(y)\, dy = 0$$

Lemma 3 (21). *With fixed x, let*

$$G(\alpha) = \int_{\mathbb{R}^m} g(\alpha, y)\rho(y)\, dy \text{ where } g(\alpha, y) = [f(x, y) - \alpha]^+$$

Then G is a convex, continuously differentiable function, with

$$G'(\alpha) = \Psi(x, \alpha) - 1$$

~~[21]~~ [Ben: Is this whole lemma due to this earlier work? If so, the change I make here is appropriate. If not, then the cite should apply to the specific step/piece and you should be clear about it...][Hrithik: Yes. Accepting change.]

Proof. Consider a function $F(\theta)$ that has the form

$$F(\theta) = \mathbb{E}\{f(\theta)\} = \int_{\Omega} f(\theta, \omega)P(d\omega)$$

where $f(\theta)$ is a measurable random function (whose realization is denoted by $f(\theta, \omega)$ as in the above integral) defined on a probability space $(\Omega, \mathcal{F}, P)$. Let the domain of f be a convex open set $D \subset \mathbb{R}^m$ and suppose that, for every $\theta \in D$, $\mathbb{E}\{|f(\theta)|\} < \infty$. We now give sufficient conditions for directional differentiability of F at a point $\theta_0 \in D$.

Assumption 2.1: There exists a positive-valued random variable $K = K(\omega)$ such that $\mathbb{E}\{K\}$ is finite and

$$|f(\theta_1, \omega) - f(\theta_2, \omega)| \leq K(\omega)\|\theta_1 - \theta_2\| \tag{2.2}$$

for almost all $\omega \in \Omega$ and for all $\theta_1, \theta_2 \in D$.

Assumption 2.2: With probability one (w.p.1) the function $f(\theta)$ is directionally differentiable at θ_0.

Proposition 2.1: Suppose that either i) Assumptions 2.1 and 2.2 hold, or ii) the function $f(\theta)$ is convex with probability 1; Then the expected-value function $F(\theta)$ is directionally differentiable at $\theta_0 \in D$ and

$$F'(\theta_0, d) = \mathbb{E}\{f'(\theta_0, d)\}. \tag{2.3}$$

Proof: Under Assumptions 2.1 and 2.2, formula (2.3) follows easily from the Lebesgue Dominated Convergence Theorem. In the convex case, formula (2.3) is implied by the Monotone Convergence Theorem. Indeed, consider $\theta_0 \in D$, $d \in \mathbb{R}^m$, and a monotone-decreasing sequence $t_n \to 0^+$. It follows then from convexity of $f(\cdot, \omega)$, that the sequence $\psi_n = \psi_n(\omega) = t_n^{-1}[f(\theta_0 + t_n d, \omega) - f(\theta_0, \omega)]$ is monotone decreasing for almost every $\omega \in \Omega$, and that $\psi_n \to f'(\theta_0, d)$ w.p.1. Therefore by the Monotone Convergence Theorem

$$\lim_{n\to\infty} \mathbb{E}\{\psi_n\} = \mathbb{E}\left\{\lim_{n\to\infty} \psi_n\right\} = \mathbb{E}\{f'(\theta_0, d)\}.$$

Since the expected-value function $F(\theta) = \mathbb{E}\{f(\theta)\}$ is then convex, we have that the limit $\lim_{n\to\infty} \mathbb{E}\{\psi_n\}$ is finite and is equal to $F'(\theta_0, d)$. This shows that, in the convex case, if $\mathbb{E}\{|f(\theta)|\} < \infty$ for all $\theta \in D$, then the interchangeability formula (2.3) holds.

Considering,

$$G(\alpha) = \mathbb{E}\{g(\alpha, y)\} \implies G'(\alpha) = \mathbb{E}\{g'(\alpha, y)\} = \Psi(x, \alpha) - 1$$

completes the proof of Lemma 01. [Ben: *I think you mean lemma 3 not 1 – but in any case you should be doing this with a latex ref statement.*]

□

With $F_\beta(\mathbf{x}, \alpha)$ as defined above, it is immediate from the Lemma 3 that $F_\beta(\mathbf{x}, \alpha)$ is convex and continuously differentiable with derivative

$$\frac{\partial}{\partial \alpha} F_\beta(\mathbf{x}, \alpha) = 1 + (1-\beta)^{-1}[\Psi(\mathbf{x}, \alpha) - 1] = (1-\beta)^{-1}[\Psi(\mathbf{x}, \alpha) - \beta].$$

Therefore, the values of α that furnish the minimum of $F_\beta(\mathbf{x}, \alpha)$, i.e., the ones comprising the set $A_\beta(\mathbf{x}) = \operatorname{argmin} F_\beta$, are precisely those for which $\Psi(\mathbf{x}, \alpha) - \beta = 0$. They form a nonempty closed interval, inasmuch as $\Psi(\mathbf{x}, \alpha)$ is continuous and nondecreasing in α with limit 1 as $\alpha \to \infty$ and limit 0 as $\alpha \to -\infty$. This further yields the validity of the β-VaR formula. In particular, then, we have

$$\min_{\alpha \in \mathbb{R}} F_\beta(\mathbf{x}, \alpha) = F_\beta(\mathbf{x}, \alpha_\beta(\mathbf{x})) = \alpha_\beta(\mathbf{x}) + (1-\beta)^{-1} \int_{\mathbf{y} \in \mathbb{R}^m} [f(\mathbf{x}, \mathbf{y}) - \alpha_\beta(\mathbf{x})]^+ p(\mathbf{y})\, d\mathbf{y}.$$

But the integral here equals

$$\int_{f(\mathbf{x},\mathbf{y}) \geq \alpha_\beta(\mathbf{x})} [f(\mathbf{x}, \mathbf{y}) - \alpha_\beta(\mathbf{x})] p(\mathbf{y})\, d\mathbf{y} = \int_{f(\mathbf{x},\mathbf{y}) \geq \alpha_\beta(\mathbf{x})} f(\mathbf{x}, \mathbf{y}) p(\mathbf{y})\, d\mathbf{y} - \alpha_\beta(\mathbf{x}) \int_{f(\mathbf{x},\mathbf{y}) \geq \alpha_\beta(\mathbf{x})} p(\mathbf{y})\, d\mathbf{y},$$

where the first integral on the right is by definition $(1-\beta)\phi_\beta(\mathbf{x})$ and the second is $1 - \Psi(\mathbf{x}, \alpha_\beta(\mathbf{x}))$. Moreover $\Psi(\mathbf{x}, \alpha_\beta(\mathbf{x})) = \beta$. Thus,

$$\min_{\alpha \in \mathbb{R}} F_\beta(\mathbf{x}, \alpha) = \alpha_\beta(\mathbf{x}) + (1-\beta)^{-1}[(1-\beta)\phi_\beta(\mathbf{x}) - \alpha_\beta(\mathbf{x})(1-\beta)] = \phi_\beta(\mathbf{x}).$$

This confirms for β-CVaR formula and finishes the proof. □

Theorem 2 ((Rockafellar and Uryasev [19])). *Minimizing the β-CVaR of the loss associated with $\mathbf{x}$ over all $\mathbf{x} \in X$ is equivalent to minimizing $F_\beta(\mathbf{x}, \alpha)$ over all $(\mathbf{x}, \alpha) \in X \times \mathbb{R}$, in the sense that*

$$\min_{\mathbf{x} \in X} \phi_\beta(\mathbf{x}) \;=\; \min_{(\mathbf{x}, \alpha) \in X \times \mathbb{R}} F_\beta(\mathbf{x}, \alpha),$$

where moreover a pair $(\mathbf{x}^*, \alpha^*)$ achieves the second minimum if and only if $\mathbf{x}^*$ achieves the first minimum and $\alpha^* \in A_\beta(\mathbf{x}^*)$. In particular, therefore, in circumstances where the interval $A_\beta(\mathbf{x}^*)$ reduces to a single point (as is typical), the minimization of $F_\beta(\mathbf{x}, \alpha)$ over $(\mathbf{x}, \alpha) \in X \times \mathbb{R}$ produces a pair $(\mathbf{x}^*, \alpha^*)$, not necessarily unique, such that $\mathbf{x}^*$ minimizes the β-CVaR and α^* gives the corresponding β-VaR.

Furthermore, $F_\beta(\mathbf{x}, \alpha)$ is convex with respect to $(\mathbf{x}, \alpha)$, and $\phi_\beta(\mathbf{x})$ is convex with respect to $\mathbf{x}$, when $f(\mathbf{x}, \mathbf{y})$ is convex with respect to $\mathbf{x}$, in which case, if the constraints are such that X is a convex set, the joint minimization is an instance of convex programming.

It should be noted that the integral in the definition of $F_\beta(\mathbf{x}, \alpha)$ can be approximated in various ways. For example, this can be done by sampling the probability distribution of y according to its density $p(\mathbf{y})$. If the sampling generates a collection of vectors $\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_q$, then the corresponding approximation to $F_\beta(\mathbf{x}, \alpha)$ is $\tilde{F}_\beta(\mathbf{x}, \alpha) = \alpha + \frac{1}{q(1-\beta)} \sum_{k=1}^{q} [f(\mathbf{x}, \mathbf{y}_k) - \alpha]^+$.

The expression $\tilde{F}_\beta(\mathbf{x}, \alpha)$ is convex and piecewise linear with respect to α. Although it is not differentiable with respect to α, it can readily be minimised, either by line search techniques or by representation in terms of an elementary linear programming problem.

CVaR is a coherent risk measure for any confidence level $\beta \in [0, 1)$. It hence has a dual representation.

Lemma 4 (Dual representation of CVaR, [3]). *For any $\beta \in [0, 1)$,*

$$CVaR_\beta[X] \;=\; \sup_{Q \in \mathcal{Q}_\beta} \mathbb{E}_Q[X], \qquad \mathcal{Q}_\beta \;=\; \left\{ Q \ll P \,:\, \frac{dQ}{dP}(\omega) \leq \frac{1}{1-\beta} \;\forall \omega \right\}. \tag{2}$$

Here $Q \ll P$ means Q is absolutely continuous *with respect to P. This means Q places mass only where P does and dQ/dP is the Radon-Nikodym derivative.*
The supremum is attained, and a worst-case measure Q^* places mass $\frac{1}{1-\beta}\rho_\omega$ on each scenario in the upper β-tail of X.

5 Stochastic Day Ahead Clearing

In this section, we formulate the stochastic day-ahead clearing problem as a two-stage program. In the first stage, tentative allocations are declared before uncertainty is realized. In the second stage, recourse actions adjust these allocations once the uncertain parameters have been realized. We present the social welfare maximization problem solved by a central market planner (the *optimization problem*), followed by the individual utility maximization problems solved independently by each market participant (the *equilibrium problem*).

5.1 The Optimization Problem

Consider a two-stage market with n consumers and m producers, indexed by i and j respectively, who wish to commit to trading quantities of a single divisible good in the first stage, for which their values and costs, and hence their optimal quantities, will be realized in the second stage. The uncertainty in this environment is modeled by a discrete random variable with outcome space Ω (e.g., in the case of solar energy, the degree of cloud cover), whose values $\omega \in \Omega$, so-called *scenarios*, are drawn from the probability distribution $\mathcal{P}$ and realised with probability ρ_ω. These scenarios parameterize the consumers' values and the producers' costs, so that each consumer i (resp. producer j) realizes a per-scenario value $v_{i\omega}$ (resp. cost $c_{j\omega}$), along with per-scenario lower and upper bounds on consumption $\underline{x}_{i\omega}$ and $\overline{x}_{i\omega}$ (resp. production $\underline{y}_{j\omega}$ and $\overline{y}_{j\omega}$).
In the first stage of the market, before any uncertainty is resolved, a pair of tentative allocations $(\boldsymbol{x}, \boldsymbol{y}) \in \mathbb{R}^n \times \mathbb{R}^m$ for the consumers and producers, respectively, is declared. In the second stage, after the scenario has been realized, recourse actions are taken via positive and negative deviations $\boldsymbol{X} = (\boldsymbol{X}^+, \boldsymbol{X}^-)$ with $\boldsymbol{X}^+, \boldsymbol{X}^- \in \mathbb{R}^{n\times|\Omega|}$ and $\boldsymbol{Y} = (\boldsymbol{Y}^+, \boldsymbol{Y}^-)$ with $\boldsymbol{Y}^+, \boldsymbol{Y}^- \in \mathbb{R}^{m\times|\Omega|}$ from the tentative allocation, that is, agents can increase or decrease their consumption or production relative to the tentative allocations. Each consumer i (resp. producer j) then suffers an exogenous marginal penalty α_i^+ (resp. α_j^+) for positive deviations and an exogenous marginal penalty α_i^- (resp. α_j^-) for negative deviations, with $\alpha_k^+, \alpha_k^- \geq 0$ for all $k \in \{i, j\}$. This asymmetry allows the market to penalize under-supply or over-demand more heavily than the opposite.
~~We model producers and consumers with *quasi-linear utilities*, meaning consumers subtract their expenses (the price of their allocations) from their consumption values, while producers subtract their production costs from their revenue.~~We model producers and consumers with quasi-linear utilities: consumers subtract their expenses and deviation penalties from their consumption values, while producers subtract their production costs and deviation penalties from their revenue. [Ben: *Suggest you make clear in previous sentence that the penalties apply in this.*] Let $p_0 \in \mathbb{R}$ be the posted price in the first stage, and let $\boldsymbol{p} \in \mathbb{R}^{|\Omega|}$ so that $p_\omega \in \mathbb{R}$ is the posted price in the second stage under scenario ω. Agent i's (quasi-linear) utility under scenario ω is:

$$\begin{aligned} u_i(x_i, \boldsymbol{X}_i, p_0, \boldsymbol{p}; \omega) = (v_{i\omega} - p_0)x_i + (v_{i\omega} - p_\omega)(x_{i\omega}^+ - x_{i\omega}^-) \\ - (\alpha_i^+ x_{i\omega}^+ + \alpha_i^- x_{i\omega}^-) \end{aligned} \tag{3}$$

Similarly, producer j's utility is:

$$\begin{aligned} u_j(y_j, \boldsymbol{Y}_j, p_0, \boldsymbol{p}; \omega) = (p_0 - c_{j\omega})y_j + (p_\omega - c_{j\omega})(y_{j\omega}^+ - y_{j\omega}^-) \\ - (\alpha_j^+ y_{j\omega}^+ + \alpha_j^- y_{j\omega}^-) \end{aligned} \tag{4}$$

For both producers and consumers, the first term represents day-ahead trades, the second represents real-time trades, and the third represents penalties for deviation from the tentative allocation. We have thus defined a *two-stage stochastic market* $\mathcal{M} = (m, n, \Omega, \mathcal{P})$ for a single divisible good, where agents' values and costs are non-negative real values, and their utilities are quasi-linear.

5.1.1 Second-Stage Welfare Problem. Given such a stochastic market $\mathcal{M}$, together with a tentative first-stage allocation $(\boldsymbol{x}, \boldsymbol{y})$, the welfare maximization problem $\mathrm{W2P}(\boldsymbol{x}, \boldsymbol{y}; \omega)$ in the second stage assuming scenario $\omega \in \Omega$ is:

$$\begin{aligned}\mathrm{W2P}(\boldsymbol{x}, \boldsymbol{y}; \omega) \;:=\; \arg\max_{X,Y} \quad & \sum_{i\in[n]} \Big[v_{i\omega}(x_i + x^+_{i\omega} - x^-_{i\omega}) - (\alpha^+_i\, x^+_{i\omega} + \alpha^-_i\, x^-_{i\omega}) \Big] \\ & - \sum_{j\in[m]} \Big[c_{j\omega}(y_j + y^+_{j\omega} - y^-_{j\omega}) + (\alpha^+_j\, y^+_{j\omega} + \alpha^-_j\, y^-_{j\omega}) \Big] \end{aligned} \tag{5}$$

subject to

$$\sum_{i\in[n]} (x^+_{i\omega} - x^-_{i\omega}) = \sum_{j\in[m]} (y^+_{j\omega} - y^-_{j\omega}) \tag{6}$$

$$x_i + x^+_{i\omega} - x^-_{i\omega} \in [\underline{x}_{i\omega},\ \overline{x}_{i\omega}] \quad \forall\, i \in [n] \tag{7}$$

$$y_j + y^+_{j\omega} - y^-_{j\omega} \in [\underline{y}_{j\omega},\ \overline{y}_{j\omega}] \quad \forall\, j \in [m] \tag{8}$$

$$\boldsymbol{X}, \boldsymbol{Y} \geq 0 \tag{9}$$

Constraint (6) ensures supply–demand balance in the second stage. Constraints (7)–(8) ensure feasibility of the second-stage allocations. Constraint (9) precludes negative adjustments; decreasing production, for example, requires a positive $y^-_{j\omega}$ rather than a negative $y^+_{j\omega}$.

Remark 1. *Our formulation enforces the equality constraint 6. In practice, it may be desirable to permit over-supply, for instance, to accommodate energy storage via battery systems. This can be achieved within our framework by introducing auxiliary slack variables representing stored or curtailed energy, thereby relaxing the balance to an inequality while preserving the dual structure from which equilibrium prices are derived.*

We write $\mathrm{W2}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{X}_\omega, Y_\omega; \omega)$ for the welfare function (the objective evaluated at the given arguments), and $\mathrm{W2}(\boldsymbol{x}, \boldsymbol{y}; \omega)$ for the optimal welfare in scenario ω.

5.1.2 First-Stage Welfare Problem. With the per-scenario second-stage problem thus defined, the first-stage welfare maximization problem $\mathrm{W1P}(\mathcal{P})$ is to maximize the expected welfare in the second stage:

$$\mathrm{W1P}(\mathcal{P}) \;:=\; \arg\max_{\boldsymbol{x},\boldsymbol{y}} \quad \mathop{\mathbb{E}}_{\omega\sim\mathcal{P}} \Big[\mathrm{W2}(\boldsymbol{x}, \boldsymbol{y}; \omega) \Big] \tag{10}$$

$$\text{subject to} \quad \sum_{i\in[n]} x_i = \sum_{j\in[m]} y_j \tag{11}$$

$$\boldsymbol{x}, \boldsymbol{y} \geq 0 \tag{12}$$

Constraint (11) ensures feasibility of the first-stage allocation, while constraint (12) ensures non-negative allocations. We write $\mathrm{W1}(\boldsymbol{x}, \boldsymbol{y}; \mathcal{P})$ for the objective of $\mathrm{W1P}(\mathcal{P})$ evaluated at the given arguments, and $\mathrm{W1}(\mathcal{P})$ for the optimal welfare.

Observation 1. *There exists an optimal solution to W1P($\mathcal{P}$) in which, for any scenario ω, at most one of $x^+_{i\omega}$ and $x^-_{i\omega}$ is nonzero for all consumers i, and similarly for producers.*

Proof. Suppose for contradiction that $x^+_{i\omega} > 0$ and $x^-_{i\omega} > 0$ for some consumer in some scenario. Let $\delta = \min(x^+_{i\omega}, x^-_{i\omega})$. Replacing $x^+_{i\omega}$ by $x^+_{i\omega} - \delta$ and $x^-_{i\omega}$ by $x^-_{i\omega} - \delta$ preserves feasibility (all constraints remain satisfied) and improves the objective by $\rho_\omega(\alpha_i^+ + \alpha_i^-)\delta \geq 0$. Thus, any optimal solution can be weakly improved until at most one deviation is nonzero per agent per scenario. □

5.1.3 Closed-Form Reformulation. Since Ω is finite, $\mathbb{E}_{\omega\sim\mathcal{P}}[\mathrm{W2}(\boldsymbol{x},\boldsymbol{y};\omega)] = \sum_{\omega\in\Omega}\rho_\omega \mathrm{W2}(\boldsymbol{x},\boldsymbol{y};\omega)$. Because the second-stage decision variables are independent across scenarios, we can swap the outer maximization with the inner sum, yielding the following equivalent closed-form program:

$$\mathrm{W1P}(\mathcal{P}) \;:=\; \arg\max_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{X},Y} \quad \sum_{\omega\in\Omega} \rho_\omega \Big[\mathrm{W2}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{X}_\omega,Y_\omega;\omega)\Big] \tag{13}$$

subject to

$$\sum_{i\in[n]} x_i = \sum_{j\in[m]} y_j \tag{14}$$

$$\sum_{i\in[n]} (x^+_{i\omega} - x^-_{i\omega}) = \sum_{j\in[m]} (y^+_{j\omega} - y^-_{j\omega}) \quad \forall\,\omega\in\Omega \tag{15}$$

$$x_i + x^+_{i\omega} - x^-_{i\omega} \in [\underline{x}_{i\omega},\ \overline{x}_{i\omega}] \quad \forall\, i\in[n],\ \forall\,\omega\in\Omega \tag{16}$$

$$y_j + y^+_{j\omega} - y^-_{j\omega} \in [\underline{y}_{j\omega},\ \overline{y}_{j\omega}] \quad \forall\, j\in[m],\ \forall\,\omega\in\Omega \tag{17}$$

$$\boldsymbol{x},\boldsymbol{y},\boldsymbol{X},Y \geq 0 \tag{18}$$

5.1.4 Dual of the First-Stage Welfare Problem. We now take the dual of the closed-form $\mathrm{W1P}(\mathcal{P})$. We assign dual variables $p_0 \geq 0$ to constraint (14), $p_\omega \in \mathbb{R}$ to each constraint (15), and $\boldsymbol{A}_i, \boldsymbol{B}_j \geq 0$ to the bound constraints. After scaling each dual variable by its corresponding scenario probability (i.e., $a^+_{i\omega} \to \rho_\omega a^+_{i\omega}$, etc.), we obtain the equivalent dual $\mathrm{W1D}(\mathcal{P})$:

$$\mathrm{W1D}(\mathcal{P}) \;:=\; \arg\min_{p_0,\boldsymbol{p},\boldsymbol{A},\boldsymbol{B}} \quad \mathop{\mathbb{E}}_{\omega\sim\mathcal{P}}\left[\sum_{i\in[n]} \left(\overline{x}_{i\omega}a^+_{i\omega} - \underline{x}_{i\omega}a^-_{i\omega}\right) + \sum_{j\in[m]} \left(\overline{y}_{j\omega}b^+_{j\omega} - \underline{y}_{j\omega}b^-_{j\omega}\right)\right] \tag{19}$$

subject to

$$\mathop{\mathbb{E}}_{\omega\sim\mathcal{P}}\left[a^+_{i\omega} - a^-_{i\omega}\right] \geq \mathop{\mathbb{E}}_{\omega\sim\mathcal{P}}\left[v_{i\omega} - (p_0 + p_\omega)\right] \qquad \forall\, i\in[n] \tag{20}$$

$$\mathop{\mathbb{E}}_{\omega\sim\mathcal{P}}\left[b^+_{j\omega} - b^-_{j\omega}\right] \geq \mathop{\mathbb{E}}_{\omega\sim\mathcal{P}}\left[(p_0 + p_\omega) - c_{j\omega}\right] \qquad \forall\, j\in[m] \tag{21}$$

$$a^+_{i\omega} - a^-_{i\omega} \geq (v_{i\omega} - \alpha_i^+) - p_\omega \qquad \forall\, i,\ \forall\,\omega \tag{22}$$

$$a^-_{i\omega} - a^+_{i\omega} \geq p_\omega - (v_{i\omega} + \alpha_i^-) \qquad \forall\, i,\ \forall\,\omega \tag{23}$$

$$b^+_{j\omega} - b^-_{j\omega} \geq p_\omega - (c_{j\omega} + \alpha_j^+) \qquad \forall\, j,\ \forall\,\omega \tag{24}$$

$$b^-_{j\omega} - b^+_{j\omega} \geq (c_{j\omega} - \alpha_j^-) - p_\omega \qquad \forall\, j,\ \forall\,\omega \tag{25}$$

$$\boldsymbol{A},\boldsymbol{B} \geq 0 \tag{26}$$

We also write $\mathrm{W1D}(\mathcal{P}; p_0, \boldsymbol{p})$ for the programme with the dual prices p_0 and $\boldsymbol{p}$ fixed.

Remark 2. *The dual variable p_0 corresponds to the first-stage balance constraint and p_ω to each second-stage balance constraint. As we show in Section 6, equilibrium prices can be read off from these dual variables.*

5.2 The Equilibrium Problem

Assuming that all agents in the market are price-takers and given prices $(p_0, \boldsymbol{p}) \in \mathbb{R} \times \mathbb{R}^{|\Omega|}$, scenario ω, and day-ahead allocations (x_i, y_j), we now define the first and second-stage utility maximisation problems for individual agents.

$$\begin{aligned} \mathrm{U2P}_i(x_i, p_0, \boldsymbol{p}; \omega) \;:=\; \arg\max_{\boldsymbol{X}_i} \quad & u_i(x_i, \boldsymbol{X}_i, p_0, \boldsymbol{p}; \omega) \\ \text{subject to} \quad & x_i + x^+_{i\omega} - x^-_{i\omega} \in [\underline{x}_{i\omega},\ \overline{x}_{i\omega}] \\ & \boldsymbol{X}_i \geq 0 \end{aligned} \tag{27}$$

$$\begin{aligned} \mathrm{U2P}_j(y_j, p_0, \boldsymbol{p}; \omega) \;:=\; \arg\max_{\boldsymbol{Y}_j} \quad & u_j(y_j, \boldsymbol{Y}_j, p_0, \boldsymbol{p}; \omega) \\ \text{subject to} \quad & y_j + y^+_{j\omega} - y^-_{j\omega} \in [\underline{y}_{j\omega},\ \overline{y}_{j\omega}] \\ & \boldsymbol{Y}_j \geq 0 \end{aligned} \tag{28}$$

We write $\mathrm{U2}_i(x_i, p_0, \boldsymbol{p}; \omega)$ and $\mathrm{U2}_j(y_j, p_0, \boldsymbol{p}; \omega)$ for the corresponding optimal values.

5.2.1 First-Stage Utility Problems. Given these, we construct the first-stage utility problems, in which each agent chooses their day-ahead allocation to maximise expected utility, anticipating optimal adjustment in the second stage:

$$\begin{aligned} \mathrm{U1P}_i(\mathcal{P}, p_0, \boldsymbol{p}) \;:=\; \arg\max_{x_i} \quad & \mathop{\mathbb{E}}_{\omega \sim \mathcal{P}} \left[\mathrm{U2}_i(x_i, p_0, \boldsymbol{p}; \omega) \right] \\ \text{subject to} \quad & x_i \geq 0 \end{aligned} \tag{29}$$

$$\begin{aligned} \mathrm{U1P}_j(\mathcal{P}, p_0, \boldsymbol{p}) \;:=\; \arg\max_{y_j} \quad & \mathop{\mathbb{E}}_{\omega \sim \mathcal{P}} \left[\mathrm{U2}_j(y_j, p_0, \boldsymbol{p}; \omega) \right] \\ \text{subject to} \quad & y_j \geq 0 \end{aligned} \tag{30}$$

We denote the optimal values by $\mathrm{U1}_i(\mathcal{P}, p_0, \boldsymbol{p})$ and $\mathrm{U1}_j(\mathcal{P}, p_0, \boldsymbol{p})$.

5.2.2 Duals of the First-Stage Utility Problems. By writing the closed-form primal (combining first and second stages) and scaling dual variables in the same manner as for W1D, we obtain the following dual programmes:

$$\mathrm{U1D}_i(\mathcal{P}, p_0, \boldsymbol{p}) \;:=\; \arg\min_{\boldsymbol{A}_i} \quad \mathop{\mathbb{E}}_{\omega \sim \mathcal{P}} \left[\overline{x}_{i\omega} a^+_{i\omega} - \underline{x}_{i\omega} a^-_{i\omega} \right] \tag{31}$$

subject to

$$\mathop{\mathbb{E}}_{\omega \sim \mathcal{P}} \left[a^+_{i\omega} - a^-_{i\omega} \right] \geq \mathop{\mathbb{E}}_{\omega \sim \mathcal{P}} \left[v_{i\omega} - p_0 \right] \tag{32}$$

$$a^+_{i\omega} - a^-_{i\omega} \geq (v_{i\omega} - \alpha^+_i) - p_\omega \quad \forall\, \omega \in \Omega \tag{33}$$

$$a^-_{i\omega} - a^+_{i\omega} \geq p_\omega - (v_{i\omega} + \alpha^-_i) \quad \forall\, \omega \in \Omega \tag{34}$$

$$\boldsymbol{A}_i \geq 0 \tag{35}$$

$$\mathrm{U1D}_j(\mathcal{P}, p_0, \boldsymbol{p}) \;:=\; \arg\min_{\boldsymbol{B}_j} \quad \mathop{\mathbb{E}}_{\omega \sim \mathcal{P}} \left[\overline{y}_{j\omega} b^+_{j\omega} - \underline{y}_{j\omega} b^-_{j\omega} \right] \tag{36}$$

subject to

$$\mathop{\mathbb{E}}_{\omega\sim\mathcal{P}}\left[b^+_{j\omega} - b^-_{j\omega}\right] \geq \mathop{\mathbb{E}}_{\omega\sim\mathcal{P}}\left[p_0 - c_{j\omega}\right] \tag{37}$$

$$b^+_{j\omega} - b^-_{j\omega} \geq p_\omega - (c_{j\omega} + \alpha^+_j) \quad \forall\, \omega \in \Omega \tag{38}$$

$$b^-_{j\omega} - b^+_{j\omega} \geq (c_{j\omega} - \alpha^-_j) - p_\omega \quad \forall\, \omega \in \Omega \tag{39}$$

$$\boldsymbol{B}_j \geq 0 \tag{40}$$

6 Stochastic Competitive Equilibrium

We now formally define the two-stage stochastic market $\mathcal{M}$. In the classical Arrow–Debreu framework, a competitive equilibrium is characterised by three properties: feasibility of allocations, utility maximisation by each agent at the posted prices, and market clearing (Walras' law). We generalise this notion to the stochastic two-stage setting, where Walras' law must hold across both stages. We refer to a tuple $(\boldsymbol{x}, \boldsymbol{y}, X, Y, p_0, \boldsymbol{p})$ as a *market outcome*.

Definition 2 (Stochastic Competitive Equilibrium with Recourse). *Given a stochastic market $\mathcal{M} = (m, n, \Omega, \mathcal{P})$, a market outcome $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, Y^*, p_0^*, \boldsymbol{p}^*)$ is a* stochastic competitive equilibrium with recourse *(SCER) if the following three conditions hold:*

(F) ***Feasibility.*** *The allocations satisfy the constraints of W1P($\mathcal{P}$): first-stage supply equals demand* (14)*, second-stage adjustments balance per scenario* (15)*, all bound constraints* (16)–(17) *hold, and all quantities are non-negative* (18)*.*

(UM) ***Utility Maximization.*** *Each agent's first- and second-stage allocation is utility maximizing at the given prices $(p_0^*, \boldsymbol{p}^*)$. Specifically, for each consumer $i \in [n]$:*

$$(x_i^*,\ \boldsymbol{X}_i^*) \in U1P_i(\mathcal{P}, p_0^*, \boldsymbol{p}^*),$$

and for each producer $j \in [m]$:

$$(y_j^*,\ \boldsymbol{Y}_j^*) \in U1P_j(\mathcal{P}, p_0^*, \boldsymbol{p}^*).$$

(WL) ***Walras' Law.*** *Let z_0 and z_ω denote the aggregate excess demand in the first and second stages, respectively:*

$$z_0 \ :=\ \sum_{i\in[n]} x_i^* - \sum_{j\in[m]} y_j^*, \qquad z_\omega \ :=\ \sum_{i\in[n]} (x_{i\omega}^{*+} - x_{i\omega}^{*-}) - \sum_{j\in[m]} (y_{j\omega}^{*+} - y_{j\omega}^{*-}).$$

The market value of aggregate excess demand is zero in each stage:

$$p_0^* \cdot z_0 = 0, \tag{41}$$

$$p_\omega^* \cdot z_\omega = 0, \quad \forall\, \omega \in \Omega. \tag{42}$$

Remark 3. *Conditions* (41)–(42) *are the natural generalization of the classical Walras' Law $\mathbf{p}\cdot\mathbf{z}(\mathbf{p}) = 0$ to the two-stage stochastic setting, applied independently to each market: the day-ahead market (a single good priced at p_0^*) and each scenario-contingent real-time market (priced at p_ω^*). Each condition asserts that when there is excess supply in a given market ($z < 0$), the corresponding price must be zero; equivalently, a strictly positive price implies exact market clearing.*

We now establish the existence of SCER via a strong-duality argument. The key insight is that prices supporting a competitive equilibrium can be read off from the dual of the welfare maximisation problem.

6.1 Existence of SCER

Theorem 3 (Existence of SCER). *Let $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, \boldsymbol{Y}^*)$ be a solution to W1P($\mathcal{P}$). Then there exist prices $(p_0^*, \boldsymbol{p}^*)$ such that the market outcome $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, \boldsymbol{Y}^*, p_0^*, \boldsymbol{p}^*)$ is an SCER.*

Proof. Since W1P($\mathcal{P}$) is a finite-dimensional convex program with affine constraints and a feasible point, strong duality holds. We write the KKT system for the closed-form first-stage formulation (13)–(18), assigning multipliers as follows:

- Day-ahead balance (14): multiplier $\lambda \geq 0$.
- For each $\omega \in \Omega$, second-stage balance (15): multiplier $\nu_\omega \in \mathbb{R}$.
- Consumer bounds (16): multipliers $\mu^{\mathrm{c}}_{i,\omega} \geq 0$ (upper) and $\underline{\mu}^{\mathrm{c}}_{i,\omega} \geq 0$ (lower).
- Producer bounds (17): multipliers $\mu^{\mathrm{g}}_{j,\omega} \geq 0$ (upper) and $\underline{\mu}^{\mathrm{g}}_{j,\omega} \geq 0$ (lower).
- Non-negativity (18): multipliers σ^{c}_i, σ^{g}_j, $\sigma^{\mathrm{c}+}_{i,\omega}$, $\sigma^{\mathrm{c}-}_{i,\omega}$, $\sigma^{\mathrm{g}+}_{j,\omega}$, $\sigma^{\mathrm{g}-}_{j,\omega} \geq 0$.

We define prices by:

$$p_0^* := \lambda, \qquad p^*_\omega := \frac{\nu_\omega}{\rho_\omega}, \quad \forall\, \omega \in \Omega. \tag{43}$$

Proof of Feasibility (F). This holds directly, since $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, \boldsymbol{Y}^*)$ is a feasible solution to W1P($\mathcal{P}$).
Proof of Utility Maximization (UM). We show that the planner's KKT conditions coincide with those of each individual agent's utility problem. Let $x_{i,\omega} := x_i + x^+_{i\omega} - x^-_{i\omega}$ and $y_{j,\omega} := y_j + y^+_{j\omega} - y^-_{j\omega}$.
Planner stationarity conditions. For each $i \in [n]$, $j \in [m]$, $\omega \in \Omega$:

$$\partial_{x_i} \mathcal{L}: \quad \textstyle\sum_\omega \rho_\omega v_{i\omega} - \lambda + \sum_\omega (\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}}_i = 0, \tag{44}$$

$$\partial_{x^+_{i\omega}} \mathcal{L}: \quad \rho_\omega(v_{i\omega} - \alpha^+_i) - \nu_\omega + (\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}+}_{i,\omega} = 0, \tag{45}$$

$$\partial_{x^-_{i\omega}} \mathcal{L}: \quad \rho_\omega(-v_{i\omega} - \alpha^-_i) + \nu_\omega - (\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}-}_{i,\omega} = 0, \tag{46}$$

$$\partial_{y_j} \mathcal{L}: \quad -\textstyle\sum_\omega \rho_\omega c_{j\omega} + \lambda + \sum_\omega (\mu^{\mathrm{g}}_{j,\omega} - \underline{\mu}^{\mathrm{g}}_{j,\omega}) - \sigma^{\mathrm{g}}_j = 0, \tag{47}$$

$$\partial_{y^+_{j\omega}} \mathcal{L}: \quad \rho_\omega(-c_{j\omega} - \alpha^+_j) + \nu_\omega + (\mu^{\mathrm{g}}_{j,\omega} - \underline{\mu}^{\mathrm{g}}_{j,\omega}) - \sigma^{\mathrm{g}+}_{j,\omega} = 0, \tag{48}$$

$$\partial_{y^-_{j\omega}} \mathcal{L}: \quad \rho_\omega(c_{j\omega} - \alpha^-_j) - \nu_\omega - (\mu^{\mathrm{g}}_{j,\omega} - \underline{\mu}^{\mathrm{g}}_{j,\omega}) - \sigma^{\mathrm{g}-}_{j,\omega} = 0. \tag{49}$$

All primal constraints (14)–(18) hold, dual feasibility requires all multipliers to be non-negative, and the standard complementary slackness relations hold for each constraint.

Agent stationarity conditions. Fix prices $(p_0^*, \boldsymbol{p}^*)$ as defined in (43) and consider consumer i maximising expected utility (3) over $(x_i, \boldsymbol{X}_i)$ subject to the per-scenario feasibility $x_{i,\omega} \in [\underline{x}_{i\omega}, \overline{x}_{i\omega}]$ and non-negativity. Let the consumer's bound and non-negativity multipliers be exactly the same symbols as above. The consumer's KKT stationarity conditions are:

$$\textstyle\sum_\omega \rho_\omega v_{i\omega} - \sum_\omega \rho_\omega p_0^* + \sum_\omega (\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}}_i = 0, \tag{50}$$

$$\rho_\omega(v_{i\omega} - p^*_\omega - \alpha^+_i) + (\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}+}_{i,\omega} = 0, \tag{51}$$

$$\rho_\omega(-v_{i\omega} + p^*_\omega - \alpha^-_i) - (\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}-}_{i,\omega} = 0. \tag{52}$$

Similarly, producer j maximises expected utility (4) with the same per-scenario feasibility, yielding:

$$-\sum_\omega \rho_\omega c_{j\omega} + \sum_\omega \rho_\omega p_0^* + \sum_\omega (\mu^{\mathrm{g}}_{j,\omega} - \underline{\mu}^{\mathrm{g}}_{j,\omega}) - \sigma^{\mathrm{g}}_j = 0, \tag{53}$$

$$\rho_\omega (p_\omega^* - c_{j\omega} - \alpha_j^+) + (\mu^{\mathrm{g}}_{j,\omega} - \underline{\mu}^{\mathrm{g}}_{j,\omega}) - \sigma^{\mathrm{g}+}_{j,\omega} = 0, \tag{54}$$

$$\rho_\omega (-p_\omega^* + c_{j\omega} - \alpha_j^-) - (\mu^{\mathrm{g}}_{j,\omega} - \underline{\mu}^{\mathrm{g}}_{j,\omega}) - \sigma^{\mathrm{g}-}_{j,\omega} = 0. \tag{55}$$

Matching the two systems. By construction, $\lambda = p_0^*$ and $\nu_\omega = \rho_\omega p_\omega^*$. Comparing (44) with (50): the planner has $-\lambda$ where the consumer has $-\sum_\omega \rho_\omega p_0^* = -p_0^*$, and since $\lambda = p_0^*$ these coincide. Comparing (45) with (51): the planner has $-\nu_\omega = -\rho_\omega p_\omega^*$ where the consumer has $-\rho_\omega p_\omega^*$, so these match identically. Likewise (46) matches (52). The producer conditions (47)–(49) match (53)–(55) by the same argument. All complementary slackness relations also coincide, since the feasible sets are identical.

Each agent's problem is convex with the same feasible region that the planner imposes, so the KKT conditions are necessary and sufficient. Therefore, at prices $(p_0^*, \boldsymbol{p}^*)$, every agent maximises their utility at allocation $(\boldsymbol{x}^*, \boldsymbol{X}^*, \boldsymbol{y}^*, Y^*)$.

Proof of Walras' Law (WL). At the optimal primal–dual pair, the complementary slackness conditions on the balance constraints yield:

$$p_0^* \left(\sum_{j \in [m]} y_j^* - \sum_{i \in [n]} x_i^* \right) = 0, \tag{56}$$

$$p_\omega^* \left(\sum_{j \in [m]} (y_{j\omega}^{*+} - y_{j\omega}^{*-}) - \sum_{i \in [n]} (x_{i\omega}^{*+} - x_{i\omega}^{*-}) \right) = 0, \quad \forall\, \omega \in \Omega. \tag{57}$$

Condition (56) is precisely Walras' law in the first stage (41). Condition (57) is precisely Walras' law in the second stage (42), holding for each scenario $\omega \in \Omega$ individually. □

Remark 4 (Price Interpretation). *The price formulas* (43) *admit a natural economic interpretation. Complementary slackness on the day-ahead balance* (14) *implies that if the constraint is slack then $p_0^* = 0$, and if it binds then p_0^* is the common marginal value/cost of day-ahead quantities. When no bound constraints are binding, the stationarity conditions* (45)–(49) *yield the familiar marginal-equals-price relations. In particular, for any scenario ω:*

$$p_\omega^* = v_{i\omega} - \alpha_i^+ \quad (\text{if } x_{i\omega}^{*+} > 0), \qquad p_\omega^* = c_{j\omega} + \alpha_j^+ \quad (\text{if } y_{j\omega}^{*+} > 0).$$

When bound constraints bind, these equalities become the corresponding one-sided conditions; equivalently, p_ω^* lies in the interval consistent with the active set at ω.

7 Fundamental Welfare Theorems of Economics

Having established the existence of SCER via Theorem 3, we now state the two fundamental welfare theorems for our stochastic two-stage market. The first welfare theorem states that any SCER allocation is welfare maximizing. The second welfare theorem asserts the converse: any welfare-maximizing allocation can be supported as an SCER by suitable prices. Together, they establish the equivalence between competitive equilibria and Pareto (social welfare) optima.

Theorem 4 (First Welfare Theorem for SCER). *If $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, \boldsymbol{Y}^*, p_0^*, \boldsymbol{p}^*)$ is an SCER, then $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, \boldsymbol{Y}^*)$ is a welfare-maximizing allocation; i.e., it solves W1P($\mathcal{P}$).*

Proof. Since the market outcome is an SCER, condition **(UM)** gives that each consumer i and producer j is individually utility maximizing at prices $(p_0^*, \boldsymbol{p}^*)$. Let $\mathcal{A}$ be the program obtained by summing all agents' utility-maximization objectives and aggregating their individual constraints.

Since the agents' problems are independent (prices are fixed), $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, \boldsymbol{Y}^*)$ is an optimal solution to $\mathcal{A}$.

We show that $\mathcal{A}$ and W1P($\mathcal{P}$) have the same solutions. By Feasibility **(F)**, the price-dependent terms in the *objective* cancel when summing agent utilities, so the objective of $\mathcal{A}$ reduces to the welfare function W1. For the *constraints*: the individual bound and non-negativity constraints are exactly those of W1P($\mathcal{P}$), while the market-clearing constraints (14)–(15) hold by condition **(F)**.

Since $\mathcal{A}$ and W1P($\mathcal{P}$) share the same objective and feasible set, any optimal solution to $\mathcal{A}$ is also optimal for W1P($\mathcal{P}$). □

Theorem 5 (Second Welfare Theorem for SCER). *If $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, \boldsymbol{Y}^*)$ is a feasible welfare-maximizing allocation (i.e., it solves W1P($\mathcal{P}$)), then there exist prices $(p_0^*, \boldsymbol{p}^*)$ such that $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, \boldsymbol{Y}^*, p_0^*, \boldsymbol{p}^*)$ is an SCER.*

Proof. This is a direct consequence of Theorem 3. Equilibrium prices are constructed from the dual multipliers of W1P($\mathcal{P}$) via the identification (43). Theorem 3 establishes that all three SCER conditions, that is, feasibility, utility maximization (via KKT matching), and Walras' law (via complementary slackness) are satisfied at these prices. □

8 Risk-Aware Day Ahead Clearing

The stochastic clearing formulation of Section 5.1 maximizes *expected* welfare, which treats all scenarios symmetrically according to their probabilities. Previous work has shown that renewable generation profiles often follow a heavy-tailed distribution [11]. Hence, in practice, market operators may wish to hedge against tail events-scenarios in which welfare is particularly poor (for example, due to renewable generation shortfalls). Conditional Value-at-Risk (CVaR), introduced in Section 4, provides a coherent risk measure that quantifies the expected welfare in the worst-case tail of the distribution.

In this section, we incorporate CVaR into the day-ahead clearing problem. We first formulate the risk-aware optimization problem as a mean–CVaR objective. We then address the equilibrium problem, where a key challenge arises: the CVaR risk measure introduces scenario-specific dual multipliers that, in general, differ across agents. Following previous work, we address this by introducing an insurance (Arrow-Debreu) securities market that completes the market and aligns agents' risk assessments with the planner's.

8.1 The Optimization Problem

We consider a social planner who seeks to maximise a convex combination of expected welfare and its CVaR at confidence level $\beta \in [0, 1)$, with the trade-off controlled by a risk-aversion parameter $\varepsilon \in [0, 1]$. Denoting the per-scenario loss by $\ell(\omega) := -\mathrm{W2}(\boldsymbol{x}, \boldsymbol{y}; \omega)$, the social planner optimises:

$$\min_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{X},Y} \quad (1-\varepsilon)\ \mathop{\mathbb{E}}_{\omega\sim\mathcal{P}}\big[\ell(\omega)\big] + \varepsilon \cdot \mathrm{CVaR}_\beta\big[\ell(\omega)\big]. \tag{58}$$

When $\varepsilon = 0$, this reduces to the risk-neutral problem W1P($\mathcal{P}$). When $\varepsilon = 1$, the planner optimises pure CVaR, hedging only against the worst $(1-\beta)$-fraction of scenarios. For intermediate values, the planner trades off expected performance against tail protection.

Using the CVaR characterisation from Theorem 2 of Section 4.1 (due to [19]), we can reformulate the CVaR term via an auxiliary variable. Introducing a scalar $t \in \mathbb{R}$ and slack variables $u_\omega \geq 0$ for each $\omega \in \Omega$, we obtain the following epigraph reformulation:

$$\min_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{X},Y,t,\{u_\omega\}} \quad -(1-\varepsilon)\sum_{\omega\in\Omega}\rho_\omega\,\mathrm{W2}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{X}_\omega,Y_\omega;\omega) + \varepsilon\left(t + \frac{1}{1-\beta}\sum_{\omega\in\Omega}\rho_\omega\,u_\omega\right) \tag{59}$$

subject to

$$\sum_{i\in[n]} x_i = \sum_{j\in[m]} y_j \tag{60}$$

$$\sum_{i\in[n]} (x^+_{i\omega} - x^-_{i\omega}) = \sum_{j\in[m]} (y^+_{j\omega} - y^-_{j\omega}) \quad \forall\, \omega \in \Omega \tag{61}$$

$$u_\omega \geq -t - \mathrm{W2}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{X}_\omega, \boldsymbol{Y}_\omega; \omega) \quad \forall\, \omega \in \Omega \tag{62}$$

$$x_i + x^+_{i\omega} - x^-_{i\omega} \in [\underline{x}_{i\omega},\ \overline{x}_{i\omega}] \quad \forall\, i,\ \forall\, \omega \tag{63}$$

$$y_j + y^+_{j\omega} - y^-_{j\omega} \in [\underline{y}_{j\omega},\ \overline{y}_{j\omega}] \quad \forall\, j,\ \forall\, \omega \tag{64}$$

$$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{X}, \boldsymbol{Y}, u_\omega \geq 0 \tag{65}$$

8.1.1 Lagrangian and KKT Conditions. Before deriving the optimality conditions, we rewrite the minimisation problem (59)–(65) as the equivalent maximisation:

$$\max_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{X},\boldsymbol{Y},t,\{u_\omega\}} \quad (1-\varepsilon) \sum_{\omega\in\Omega} \rho_\omega\, \mathrm{W2}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{X}_\omega, \boldsymbol{Y}_\omega; \omega) - \varepsilon\left(t + \frac{1}{1-\beta} \sum_{\omega\in\Omega} \rho_\omega\, u_\omega\right) \tag{66}$$

subject to the same constraints (60)–(65). We assign dual variables as follows:

- Day-ahead balance (60): multiplier $\lambda \geq 0$.
- For each $\omega \in \Omega$, real-time balance (61): multiplier $\nu_\omega \in \mathbb{R}$.
- For each $\omega \in \Omega$, CVaR shortfall (62): multiplier $\gamma_\omega \geq 0$.
- For each $\omega \in \Omega$, non-negativity $u_\omega \geq 0$: multiplier $\eta_\omega \geq 0$.
- Consumer bounds (63): multipliers $\mu^{\mathrm{c}}_{i,\omega} \geq 0$ (upper) and $\underline{\mu}^{\mathrm{c}}_{i,\omega} \geq 0$ (lower).
- Producer bounds (64): multipliers $\mu^{\mathrm{g}}_{j,\omega} \geq 0$ (upper) and $\underline{\mu}^{\mathrm{g}}_{j,\omega} \geq 0$ (lower).
- Non-negativity (65): multipliers σ^{c}_i, σ^{g}_j, $\sigma^{\mathrm{c}+}_{i,\omega}$, $\sigma^{\mathrm{c}-}_{i,\omega}$, $\sigma^{\mathrm{g}+}_{j,\omega}$, $\sigma^{\mathrm{g}-}_{j,\omega} \geq 0$.

We define the *CVaR-tilted probability weight*

$$\tilde{\rho}_\omega \ :=\ (1-\varepsilon)\,\rho_\omega + \gamma_\omega, \qquad \forall\, \omega \in \Omega. \tag{67}$$

Let $x_{i,\omega} := x_i + x^+_{i\omega} - x^-_{i\omega}$ and $y_{j,\omega} := y_j + y^+_{j\omega} - y^-_{j\omega}$. The Lagrangian is:

$$\begin{aligned}
\mathcal{L} = \sum_{\omega\in\Omega} \tilde{\rho}_\omega \Bigg[& \sum_{i\in[n]} \left(v_{i\omega}\, x_{i,\omega} - \alpha^+_i\, x^+_{i\omega} - \alpha^-_i\, x^-_{i\omega} \right) \\
& \qquad - \sum_{j\in[m]} \left(c_{j\omega}\, y_{j,\omega} + \alpha^+_j\, y^+_{j\omega} + \alpha^-_j\, y^-_{j\omega} \right) \Bigg] \\
& + \Big(\sum_{\omega\in\Omega} \gamma_\omega - \varepsilon \Big)\, t + \sum_{\omega\in\Omega} \left(\gamma_\omega + \eta_\omega - \frac{\varepsilon\, \rho_\omega}{1-\beta} \right) u_\omega \\
& + \lambda \Big(\sum_{j\in[m]} y_j - \sum_{i\in[n]} x_i \Big) + \sum_{\omega\in\Omega} \nu_\omega \Big(\sum_{j\in[m]} (y^+_{j\omega} - y^-_{j\omega}) - \sum_{i\in[n]} (x^+_{i\omega} - x^-_{i\omega}) \Big) \\
& + \sum_{\omega\in\Omega} \sum_{i\in[n]} \left[\mu^{\mathrm{c}}_{i,\omega}\, (x_{i,\omega} - \overline{x}_{i\omega}) - \underline{\mu}^{\mathrm{c}}_{i,\omega}\, (x_{i,\omega} - \underline{x}_{i\omega}) \right] \\
& + \sum_{\omega\in\Omega} \sum_{j\in[m]} \left[\mu^{\mathrm{g}}_{j,\omega}\, (y_{j,\omega} - \overline{y}_{j\omega}) - \underline{\mu}^{\mathrm{g}}_{j,\omega}\, (y_{j,\omega} - \underline{y}_{j\omega}) \right] \\
& - \sum_{i\in[n]} \sigma^{\mathrm{c}}_i\, x_i - \sum_{j\in[m]} \sigma^{\mathrm{g}}_j\, y_j \\
& - \sum_{\omega\in\Omega} \sum_{i\in[n]} \left(\sigma^{\mathrm{c}+}_{i,\omega}\, x^+_{i\omega} + \sigma^{\mathrm{c}-}_{i,\omega}\, x^-_{i\omega} \right) - \sum_{\omega\in\Omega} \sum_{j\in[m]} \left(\sigma^{\mathrm{g}+}_{j,\omega}\, y^+_{j\omega} + \sigma^{\mathrm{g}-}_{j,\omega}\, y^-_{j\omega} \right).
\end{aligned} \tag{68}$$

8.1.2 KKT Stationarity Conditions. Setting $\partial_t \mathcal{L} = 0$ and $\partial_{u_\omega} \mathcal{L} = 0$ yields:

$$\partial_t \mathcal{L} : \quad \sum_{\omega\in\Omega} \gamma_\omega = \varepsilon, \tag{69}$$

$$\partial_{u_\omega} \mathcal{L} : \quad \gamma_\omega + \eta_\omega = \frac{\varepsilon\, \rho_\omega}{1-\beta}, \quad \forall\, \omega \in \Omega. \tag{70}$$

Since $\eta_\omega \geq 0$, it follows that $0 \leq \gamma_\omega \leq \varepsilon\, \rho_\omega/(1-\beta)$ for all ω. Moreover, $\sum_\omega \tilde{\rho}_\omega = 1 + \varepsilon$, so the normalised weights $\tilde{\rho}_\omega/(1+\varepsilon)$ define a probability measure that places additional mass on tail scenarios (those with $\gamma_\omega > 0$).

Connecting back to Lemma 4: γ^*_ω is precisely the multiplier identifying the worst-case measure in CVaR's dual representation (2), and the bound $0 \leq \gamma_\omega \leq \varepsilon\rho_\omega/(1-\beta)$ derived above is exactly the density bound $dQ/dP \leq 1/(1-\beta)$ from (2), rescaled by ε. The planner's KKT system thus encodes precisely the probability distortion induced by tail-aversion on the multiplier for the CVaR shortfall constraint (62) which is a fact we use in the price identifications below.

The stationarity conditions for the physical variables are identical in structure to the risk-neutral conditions (44)–(49) with ρ_ω replaced everywhere by $\tilde{\rho}_\omega$:

$$\partial_{x_i}\mathcal{L}: \quad \sum_\omega \tilde{\rho}_\omega\, v_{i\omega} - \lambda + \sum_\omega(\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}}_i = 0, \tag{71}$$

$$\partial_{x^+_{i\omega}}\mathcal{L}: \quad \tilde{\rho}_\omega(v_{i\omega} - \alpha^+_i) - \nu_\omega + (\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}+}_{i,\omega} = 0, \tag{72}$$

$$\partial_{x^-_{i\omega}}\mathcal{L}: \quad \tilde{\rho}_\omega(-v_{i\omega} - \alpha^-_i) + \nu_\omega - (\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}-}_{i,\omega} = 0, \tag{73}$$

$$\partial_{y_j}\mathcal{L}: \quad -\sum_\omega \tilde{\rho}_\omega\, c_{j\omega} + \lambda + \sum_\omega(\mu^{\mathrm{g}}_{j,\omega} - \underline{\mu}^{\mathrm{g}}_{j,\omega}) - \sigma^{\mathrm{g}}_j = 0, \tag{74}$$

$$\partial_{y^+_{j\omega}}\mathcal{L}: \quad \tilde{\rho}_\omega(-c_{j\omega} - \alpha^+_j) + \nu_\omega + (\mu^{\mathrm{g}}_{j,\omega} - \underline{\mu}^{\mathrm{g}}_{j,\omega}) - \sigma^{\mathrm{g}+}_{j,\omega} = 0, \tag{75}$$

$$\partial_{y^-_{j\omega}}\mathcal{L}: \quad \tilde{\rho}_\omega(c_{j\omega} - \alpha^-_j) - \nu_\omega - (\mu^{\mathrm{g}}_{j,\omega} - \underline{\mu}^{\mathrm{g}}_{j,\omega}) - \sigma^{\mathrm{g}-}_{j,\omega} = 0. \tag{76}$$

All primal constraints hold, dual feasibility requires all multipliers to be non-negative, and the standard complementary slackness relations hold for each constraint, together with

$$\gamma_\omega\big(u_\omega + \mathrm{W2}(\cdot\,;\omega) + t\big) = 0, \qquad \eta_\omega\, u_\omega = 0, \qquad \forall\,\omega \in \Omega. \tag{77}$$

8.2 The Equilibrium Problem

We now define the risk-aware utility maximization problem for individual agents. Each agent evaluates their quasi-linear utility using the same mean–CVaR criterion as the social planner, with common risk-aversion parameter ε and confidence level β. Assuming that individual agents share the planner's risk-aversion parameter ε and confidence level β is a strong assumption. We adopt the common-parameter assumption here as a starting point because it isolates the question of whether risk-averse preferences, even when shared, can be decentralised as a competitive equilibrium at all. Then we move on to asking how heterogeneous preferences complicate the picture. As we show in Section 8.2.2, decentralisation already fails under common parameters unless agents' CVaR tail multipliers coincide. This motivates the insurance market that we introduce in Section 8.3. We return to the case of agents having hetrogenous risk preferences in Section 8.7, where we characterize precisely which forms of heterogeneity in (ε, β) can still be supported in equilibrium.

Given posted prices $(p_0, \boldsymbol{p})$, consumer i solves the following first-stage problem:

[Ben: *The assumption that the agents have the same risk aversion as the central planner is* extremely *strong. In general I would expect individual agents to be risk neutral, as corporate entities normally approximately are, and the central planner to be risk averse. I suggest you anticipate the reader having this reaction and position around it in some way here. Indeed the whole section is effectively leading to this conclusion... but you need to foreshadow this here because the reader thinks you aren't anticipating this problem. Make clear that you will shortly be allowing the agents more power – that of buying/selling insurance – which helps them mitigate their risk – but which does not fix this homogeneity problem. Then explicitly point forward to where you will discuss it.*]

$$\begin{aligned}\mathrm{CVaRP}_i(\mathcal{P}, p_0, \boldsymbol{p}) \;:=\; \arg\max_{x_i, \boldsymbol{X}_i, t^c_i, \{\zeta^c_{i,\omega}\}} \quad & (1-\varepsilon)\sum_{\omega\in\Omega}\rho_\omega\, u_i(x_i, \boldsymbol{X}_i, p_0, \boldsymbol{p}; \omega) \\ & - \varepsilon\left(t^c_i + \frac{1}{1-\beta}\sum_{\omega\in\Omega}\rho_\omega\,\zeta^c_{i,\omega}\right)\end{aligned} \tag{78}$$

subject to

$$\zeta^c_{i,\omega} \geq -u_i(x_i, \boldsymbol{X}_i, p_0, \boldsymbol{p}; \omega) - t^c_i, \quad \forall\, \omega \in \Omega \tag{79}$$

$$x_i + x^+_{i\omega} - x^-_{i\omega} \in [\underline{x}_{i\omega},\ \overline{x}_{i\omega}], \quad \forall\, \omega \in \Omega \tag{80}$$

$$x_i, \boldsymbol{X}_i, \zeta^c_{i,\omega} \geq 0 \tag{81}$$

where u_i is the quasi-linear utility defined in (3). The agent maximizes expected utility less ε times the CVaR of their negative utility (loss), hedging against scenarios in which their utility is particularly poor.

Similarly, producer j solves:

$$\begin{aligned}\mathrm{CVaRP}_j(\mathcal{P}, p_0, \boldsymbol{p}) \;:=\; \arg\max_{y_j, \boldsymbol{Y}_j, t^g_j, \{\zeta^g_{j,\omega}\}} \quad & (1-\varepsilon) \sum_{\omega\in\Omega} \rho_\omega\, u_j(y_j, \boldsymbol{Y}_j, p_0, \boldsymbol{p}; \omega) \\ & - \varepsilon\left(t^g_j + \frac{1}{1-\beta} \sum_{\omega\in\Omega} \rho_\omega\, \zeta^g_{j,\omega}\right)\end{aligned} \tag{82}$$

subject to

$$\zeta^g_{j,\omega} \geq -u_j(y_j, \boldsymbol{Y}_j, p_0, \boldsymbol{p}; \omega) - t^g_j, \quad \forall\, \omega \in \Omega \tag{83}$$

$$y_j + y^+_{j\omega} - y^-_{j\omega} \in [\underline{y}_{j\omega},\ \overline{y}_{j\omega}], \quad \forall\, \omega \in \Omega \tag{84}$$

$$y_j, \boldsymbol{Y}_j, \zeta^g_{j,\omega} \geq 0 \tag{85}$$

8.2.1 Lagrangian of the Equilibrium Problem. We now write the Lagrangian for the consumer's risk-aware problem (78)–(81). Introduce dual variables $\gamma^c_{i,\omega} \geq 0$ for each CVaR shortfall constraint (79), $\eta^c_{i,\omega} \geq 0$ for non-negativity of $\zeta^c_{i,\omega}$, and the usual bound/non-negativity multipliers $\mu^{\mathrm{c}}_{i,\omega}$, $\underline{\mu}^{\mathrm{c}}_{i,\omega}$, σ^{c}_i, $\sigma^{\mathrm{c}+}_{i,\omega}$, $\sigma^{\mathrm{c}-}_{i,\omega} \geq 0$. The Lagrangian is:

$$\begin{aligned}\mathcal{L}^{\mathrm{c}}_i = & \sum_{\omega\in\Omega} \left((1-\varepsilon)\,\rho_\omega + \gamma^c_{i,\omega}\right) u_i(x_i, \boldsymbol{X}_i, p_0, \boldsymbol{p}; \omega) \\ & + \left(\sum_{\omega\in\Omega} \gamma^c_{i,\omega} - \varepsilon\right) t^c_i + \sum_{\omega\in\Omega} \left(\gamma^c_{i,\omega} + \eta^c_{i,\omega} - \frac{\varepsilon\,\rho_\omega}{1-\beta}\right) \zeta^c_{i,\omega} \\ & + \sum_{\omega\in\Omega} \left[\mu^{\mathrm{c}}_{i,\omega}\,(x_{i,\omega} - \overline{x}_{i\omega}) - \underline{\mu}^{\mathrm{c}}_{i,\omega}\,(x_{i,\omega} - \underline{x}_{i\omega})\right] \\ & - \sigma^{\mathrm{c}}_i\, x_i - \sum_{\omega\in\Omega} \left(\sigma^{\mathrm{c}+}_{i,\omega}\, x^+_{i\omega} + \sigma^{\mathrm{c}-}_{i,\omega}\, x^-_{i\omega}\right).\end{aligned} \tag{86}$$

Setting $\partial_{t^c_i} \mathcal{L}^{\mathrm{c}}_i = 0$ and $\partial_{\zeta^c_{i,\omega}} \mathcal{L}^{\mathrm{c}}_i = 0$ gives the agent-specific analogues of (69)–(70):

$$\sum_{\omega\in\Omega} \gamma^c_{i,\omega} = \varepsilon, \qquad \gamma^c_{i,\omega} + \eta^c_{i,\omega} = \frac{\varepsilon\,\rho_\omega}{1-\beta}, \quad \forall\, \omega \in \Omega. \tag{87}$$

Defining the agent's tilted weight $\tilde{\rho}^c_{i,\omega} := (1-\varepsilon)\,\rho_\omega + \gamma^c_{i,\omega}$, we have $\sum_\omega \tilde{\rho}^c_{i,\omega} = (1-\varepsilon) + \varepsilon = 1$, so the agent's tilted weights also form a probability measure. The stationarity conditions for x_i, $x^+_{i\omega}$, and $x^-_{i\omega}$ become:

$$\partial_{x_i} \mathcal{L}_i^{\mathrm{c}} : \quad \sum_{\omega} \tilde{\rho}_{i,\omega}^{c} \, (v_{i\omega} - p_0) + \sum_{\omega} (\mu_{i,\omega}^{\mathrm{c}} - \underline{\mu}_{i,\omega}^{\mathrm{c}}) - \sigma_i^{\mathrm{c}} = 0, \tag{88}$$

$$\partial_{x_{i\omega}^{+}} \mathcal{L}_i^{\mathrm{c}} : \quad \tilde{\rho}_{i,\omega}^{c} \, (v_{i\omega} - p_\omega - \alpha_i^{+}) + (\mu_{i,\omega}^{\mathrm{c}} - \underline{\mu}_{i,\omega}^{\mathrm{c}}) - \sigma_{i,\omega}^{\mathrm{c}+} = 0, \tag{89}$$

$$\partial_{x_{i\omega}^{-}} \mathcal{L}_i^{\mathrm{c}} : \quad \tilde{\rho}_{i,\omega}^{c} \, (-v_{i\omega} + p_\omega - \alpha_i^{-}) - (\mu_{i,\omega}^{\mathrm{c}} - \underline{\mu}_{i,\omega}^{\mathrm{c}}) - \sigma_{i,\omega}^{\mathrm{c}-} = 0. \tag{90}$$

The producer's Lagrangian and KKT conditions are analogous, with tilted weight $\tilde{\rho}_{j,\omega}^{g} := (1 - \varepsilon)\, \rho_\omega + \gamma_{j,\omega}^{g}$ (also summing to 1) and stationarity:

$$\partial_{y_j} \mathcal{L}_j^{\mathrm{g}} : \quad -\sum_{\omega} \tilde{\rho}_{j,\omega}^{g} \, (c_{j\omega} - p_0) + \sum_{\omega} (\mu_{j,\omega}^{\mathrm{g}} - \underline{\mu}_{j,\omega}^{\mathrm{g}}) - \sigma_j^{\mathrm{g}} = 0, \tag{91}$$

$$\partial_{y_{j\omega}^{+}} \mathcal{L}_j^{\mathrm{g}} : \quad \tilde{\rho}_{j,\omega}^{g} \, (p_\omega - c_{j\omega} - \alpha_j^{+}) + (\mu_{j,\omega}^{\mathrm{g}} - \underline{\mu}_{j,\omega}^{\mathrm{g}}) - \sigma_{j,\omega}^{\mathrm{g}+} = 0, \tag{92}$$

$$\partial_{y_{j\omega}^{-}} \mathcal{L}_j^{\mathrm{g}} : \quad \tilde{\rho}_{j,\omega}^{g} \, (-p_\omega + c_{j\omega} - \alpha_j^{-}) - (\mu_{j,\omega}^{\mathrm{g}} - \underline{\mu}_{j,\omega}^{\mathrm{g}}) - \sigma_{j,\omega}^{\mathrm{g}-} = 0. \tag{93}$$

8.2.2 Challenge for decentralization. We now compare the planner's KKT (71)–(76) with the agents' KKT (88)–(93). Consider the stationarity condition for x_i. The planner (71) has:

$$\sum_{\omega} \tilde{\rho}_\omega \, v_{i\omega} - \lambda + \cdots = 0,$$

while the agent (88) has:

$$\sum_{\omega} \tilde{\rho}_{i,\omega}^{c} \, v_{i\omega} - p_0 + \cdots = 0,$$

since $\sum_\omega \tilde{\rho}_{i,\omega}^{c} = 1$. For the real-time variables, comparing (72) with (89) requires $\tilde{\rho}_\omega = \tilde{\rho}_{i,\omega}^{c}$ and $\nu_\omega = \tilde{\rho}_\omega \, p_\omega$. Therefore, the planner's and agents' conditions coincide if and only if:

$$\gamma_{i,\omega}^{c} = \gamma_{j,\omega}^{g} = \gamma_\omega, \qquad \forall\, i \in [n],\ j \in [m],\ \omega \in \Omega. \tag{94}$$

That is, all agents' CVaR tail multipliers must equal the planner's. In general, there is no reason for this to hold: each agent's CVaR multiplier $\gamma_{i,\omega}^{c}$ is determined by the tail structure of their *individual* utility, which varies across agents. Two agents may experience very different utility levels in the same scenario (e.g., a solar generator and a gas generator face different cost structures under the same weather outcome), leading to different sets of tail scenarios and hence different γ-vectors.

Without a mechanism to align these multipliers, the risk-averse planner's solution *cannot* in general be decentralised as a competitive equilibrium. This observation is consistent with the findings of [9], who demonstrate that in incomplete risk markets, risk-averse equilibria may be non-unique and need not coincide with the social planner's optimum. To resolve this, we introduce a market for state-contingent claims that allows agents to trade risk.

8.3 Insurance Market

[Ben: *More transition/framing here is gonna make the reader better able to connect this section to the previous, and why you are adding insurance to the market. What is this addition really do and why? Presumably it is there to help agents better cope with their risk profiles, or to profit off their ability to take on risk – just like any insurance transaction. You should be really clear here that you are adding something useful to the market, but that you are not (yet) explicitly relaxing the assumption of homogeneous risk aversion, and allowing the planner and the bidders to differ, and the players to differ*

among them selves. Then, be clear that you are not yet handling this homogeneity problem, but will soon. This may mean that you move the content of remark 6 up here... or it may mean you foreshadow here and then refine its content below. In either case, I think there should be some content here to keep the reader on board.]

We complete the market by introducing *Arrow–Debreu securities* [4]. For each scenario $\omega \in \Omega$, an Arrow–Debreu security is a contract that charges a price q_ω in the first stage and delivers one unit of payment in scenario ω. For each consumer i and producer j, let

$$r^c_{i\omega} \in \mathbb{R}, \qquad r^g_{j\omega} \in \mathbb{R}$$

denote the net units of the Arrow–Debreu security for scenario ω purchased by consumer i and producer j, respectively. Positive values represent insurance indemnities received; negative values represent premiums paid.

The insurance market clears state-by-state:

$$\sum_{i\in[n]} r^c_{i\omega} + \sum_{j\in[m]} r^g_{j\omega} = 0, \qquad \forall\, \omega \in \Omega. \tag{95}$$

8.4 The Equilibrium Problem with Insurance

With the insurance market available, each agent's problem is augmented with insurance decisions. Consumer i solves:

$$\max_{x_i, \boldsymbol{X}_i, \{r^c_{i\omega}\}, t^c_i, \{\zeta^c_{i,\omega}\}} \quad (1-\varepsilon) \sum_{\omega\in\Omega} \rho_\omega \left[u_i(x_i, \boldsymbol{X}_i, p_0, \boldsymbol{p}; \omega) + r^c_{i\omega}\right] - \sum_{\omega\in\Omega} q_\omega\, r^c_{i\omega} - \varepsilon\left(t^c_i + \frac{1}{1-\beta} \sum_{\omega\in\Omega} \rho_\omega\, \zeta^c_{i,\omega}\right) \tag{96}$$

subject to

$$\zeta^c_{i,\omega} \geq -u_i(x_i, \boldsymbol{X}_i, p_0, \boldsymbol{p}; \omega) - r^c_{i\omega} - t^c_i, \quad \forall\, \omega \in \Omega \tag{97}$$

$$x_i + x^+_{i\omega} - x^-_{i\omega} \in [\underline{x}_{i\omega}, \overline{x}_{i\omega}], \quad \forall\, \omega \in \Omega \tag{98}$$

$$x_i, \boldsymbol{X}_i, \zeta^c_{i,\omega} \geq 0. \tag{99}$$

The agent's stochastic per-scenario payoff is $u_i(\cdot\,;\omega) + r^c_{i\omega}$, that is, the energy utility plus the contingent insurance payout in scenario ω. Risk preferences are applied to this stochastic payoff, yielding the $(1-\varepsilon)$-weighted expectation of $u_i + r^c_{i\omega}$ together with the ε-weighted CVaR of its negative (i.e., the insured loss $-u_i - r^c_{i\omega}$). The insurance premium $\sum_\omega q_\omega\, r^c_{i\omega}$, paid in the first stage before any uncertainty resolves, is a deterministic cost that sits outside the risk measure.

Similarly, producer j solves:

$$\max_{y_j, \boldsymbol{Y}_j, \{r^g_{j\omega}\}, t^g_j, \{\zeta^g_{j,\omega}\}} \quad (1-\varepsilon) \sum_{\omega\in\Omega} \rho_\omega \left[u_j(y_j, \boldsymbol{Y}_j, p_0, \boldsymbol{p}; \omega) + r^g_{j\omega}\right] - \sum_{\omega\in\Omega} q_\omega\, r^g_{j\omega} - \varepsilon\left(t^g_j + \frac{1}{1-\beta} \sum_{\omega\in\Omega} \rho_\omega\, \zeta^g_{j,\omega}\right) \tag{100}$$

subject to

$$\zeta^g_{j,\omega} \geq -u_j(y_j, Y_j, p_0, \boldsymbol{p}; \omega) - r^g_{j\omega} - t^g_j, \quad \forall\, \omega \in \Omega \tag{101}$$

$$y_j + y^+_{j\omega} - y^-_{j\omega} \in [\underline{y}_{j\omega},\ \overline{y}_{j\omega}], \quad \forall\, \omega \in \Omega \tag{102}$$

$$y_j, Y_j, \zeta^g_{j,\omega} \geq 0. \tag{103}$$

8.5 Risk-Aware Stochastic Competitive Equilibrium (R-SCER)

We now formalize the notion of competitive equilibrium for the risk-aware market augmented with insurance trading.

Definition 3 (Risk-Aware SCER with Insurance). *Given a stochastic market $\mathcal{M} = (m, n, \Omega, \mathcal{P})$ and common risk parameters (ε, β), a market outcome $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, Y^*, r^{c*}, r^{g*}, p^*_0, \boldsymbol{p}^*, q^*)$ is a* risk-aware stochastic competitive equilibrium with recourse and insurance *(R-SCER) if the following conditions hold:*

(F) ***Feasibility.*** *The physical allocations satisfy the constraints of the risk-aware planner problem* (60)–(65).

(UM) ***Utility Maximization.*** *At prices $(p^*_0, \boldsymbol{p}^*, q^*)$, each consumer i maximizes their risk-adjusted utility: $(x^*_i, \boldsymbol{X}^*_i, r^{c*}_i)$ solves* (96)–(99). *Similarly, each producer j maximizes their risk-adjusted utility: (y^*_j, Y^*_j, r^{g*}_j) solves* (100)–(103).

(WL) ***Walras' Law.*** *The market value of aggregate excess demand is zero in each market:*

$$p^*_0 \cdot z_0 = 0, \tag{104}$$

$$p^*_\omega \cdot z_\omega = 0, \qquad \forall\, \omega \in \Omega, \tag{105}$$

$$q^*_\omega \cdot \Big(\sum_{i \in [n]} r^c_{i\omega^*} + \sum_{j \in [m]} r^g_{j\omega^*} \Big) = 0, \qquad \forall\, \omega \in \Omega, \tag{106}$$

where z_0 and z_ω are the aggregate excess demands defined as in Definition 2.

Remark 5. *Condition* (106) *is the natural Walras' law for the insurance market: if there is excess supply of insurance in scenario ω, the corresponding price q^*_ω is zero. A strictly positive insurance price implies that the insurance market clears exactly in that state.*

Remark 6. *The assumption of common risk parameters (ε, β) is substantive. As discussed in Section 8.2.2, the alignment of CVaR multipliers requires all agents to agree on the risk-aversion weight ε and the confidence level β. The insurance market aligns the tail-weighting* across agents *(the γ-vectors), but it cannot reconcile fundamentally different risk preferences. We discuss the extension of the mechanism which relaxes to different risk-aversion β in ~~a later section~~Section 8.7. [Ben:* What section? include a latex ref.*]*

[Ben: This remark is getting at what I am talking about in my comment above. But given how important it is, I think you need to foreshadow it more up there, see that comment. And this comment would benefit from being stronger about what is offered in the later section to punch up that the heterogeneity / modeling realism, will indeed be addressed.*]*

8.5.1 *Existence of R-SCER..*

Theorem 6 (Existence of R-SCER). *Let $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, Y^*, t^*, u^*)$ be a solution to the risk-aware planner problem* (66) *with optimal dual variables $(\lambda^*, \nu^*_\omega, \gamma^*_\omega, \eta^*_\omega)$ and the associated bound/non-negativity multipliers. Define prices:*

$$p^*_0 := \lambda^*, \qquad p^*_\omega := \frac{\nu^*_\omega}{\tilde{\rho}^*_\omega}, \qquad q^*_\omega := \tilde{\rho}^*_\omega, \qquad \forall\, \omega \in \Omega, \tag{107}$$

where $\tilde{\rho}^*_\omega = (1-\varepsilon)\,\rho_\omega + \gamma^*_\omega$.

Then there exist insurance transfers ${r^c_{i\omega}}^*$ *and* ${r^g_{j\omega}}^*$ *such that* $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, \boldsymbol{Y}^*, r^{c*}, r^{g*}, p^*_0, \boldsymbol{p}^*, q^*)$ *is an R-SCER.*

Proof. The proof establishes each R-SCER condition in turn.

Proof of Feasibility (F). This holds directly, since $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, \boldsymbol{Y}^*)$ is a feasible solution to the planner problem.

Proof of Utility Maximization (UM). We show that the planner's KKT conditions coincide with those of each insured agent at the prices (107).

Insurance prices align CVaR multipliers. Consider the Lagrangian of consumer i's insured problem (96)–(99). Let $\gamma^c_{i,\omega} \geq 0$ denote the multiplier on the CVaR shortfall constraint (97). Stationarity with respect to $r^c_{i\omega}$ gives:

$$\partial_{r^c_{i\omega}} \mathcal{L}^{\mathrm{c}}_i = (1-\varepsilon)\rho_\omega - q_\omega + \gamma^c_{i,\omega} = 0 \implies q_\omega = (1-\varepsilon)\rho_\omega + \gamma^c_{i,\omega}. \tag{108}$$

The right-hand side is precisely the agent's CVaR-tilted weight $\tilde{\rho}^c_{i,\omega}$, so equilibrium insurance prices coincide with each agent's tilted probability mass: $q_\omega = \tilde{\rho}^c_{i,\omega}$. An analogous computation for producer j gives $q_\omega = \tilde{\rho}^g_{j,\omega}$.

Setting $q^*_\omega = \tilde{\rho}^*_\omega = (1-\varepsilon)\rho_\omega + \gamma^*_\omega$ ensures that every agent's tilted weight equals the planner's:

$$\tilde{\rho}^c_{i,\omega} = \tilde{\rho}^g_{j,\omega} = \tilde{\rho}^*_\omega, \qquad \forall\, i, j, \omega, \tag{109}$$

or equivalently, every agent's CVaR multiplier equals the planner's: $\gamma^c_{i,\omega} = \gamma^g_{j,\omega} = \gamma^*_\omega$, assuming that the agents share knowledge of the same probability distribution and ϵ.

Real-time variables. Comparing the agent's condition (89) (with $\tilde{\rho}^c_{i,\omega} = \tilde{\rho}^*_\omega$):

$$\tilde{\rho}^*_\omega(v_{i\omega} - p^*_\omega - \alpha^+_i) + (\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}+}_{i,\omega} = 0$$

with the planner's condition (72):

$$\tilde{\rho}^*_\omega(v_{i\omega} - \alpha^+_i) - \nu^*_\omega + (\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}+}_{i,\omega} = 0,$$

these coincide if and only if $\tilde{\rho}^*_\omega\, p^*_\omega = \nu^*_\omega$, which holds by (107). The conditions for $x^-_{i\omega}$, $y^+_{j\omega}$, and $y^-_{j\omega}$ match by the same argument.

Day-ahead variables. The agent's condition (88) (with $\tilde{\rho}^c_{i,\omega} = \tilde{\rho}^*_\omega$) reads:

$$\sum_\omega \tilde{\rho}^*_\omega\, v_{i\omega} - \Big(\sum_\omega \tilde{\rho}^*_\omega\Big)\, p^*_0 + \sum_\omega (\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}}_i = 0.$$

Since $\sum_\omega \tilde{\rho}^*_\omega = (1-\varepsilon) + \varepsilon = 1$, this simplifies to:

$$\sum_\omega \tilde{\rho}^*_\omega\, v_{i\omega} - p^*_0 + \sum_\omega (\mu^{\mathrm{c}}_{i,\omega} - \underline{\mu}^{\mathrm{c}}_{i,\omega}) - \sigma^{\mathrm{c}}_i = 0.$$

Comparing with the planner's condition (71), these coincide if and only if $\lambda^* = p^*_0$, which holds by (107). The producer's day-ahead condition matches analogously.

Since each agent's problem is convex with the same feasible region that the planner imposes on that agent's variables, the KKT conditions are necessary and sufficient. Therefore, at prices $(p^*_0, \boldsymbol{p}^*, q^*)$, every agent maximises their risk-adjusted utility at the planner's allocation (for any choice of insurance transfers consistent with complementary slackness). We illustrate a construction below that provides one such choice.

[Ben: *Having a remark inside a proof is undesirable. rework so the remark is outside of the proof environment.*]

Remark 7. *We construct explicit insurance transfers satisfying* (95)*. Define each agent's loss at the planner's solution:*

$$L^c_{i\omega} := -u_i(x^*_i, \boldsymbol{X}^*_i, p^*_0, \boldsymbol{p}^*; \omega), \qquad L^g_{j\omega} := -u_j(y^*_j, Y^*_j, p^*_0, \boldsymbol{p}^*; \omega).$$

By Walras' law (established below), the price-dependent terms cancel when summing over all agents:

$$\sum_{i\in[n]} L^c_{i\omega} + \sum_{j\in[m]} L^g_{j\omega} = \ell^*(\omega) := -W2(\boldsymbol{x}^*, \boldsymbol{y}^*; \omega), \qquad \forall\, \omega \in \Omega. \tag{110}$$

Choose constants δ^c_i and δ^g_j (independent of ω) satisfying $\sum_i \delta^c_i + \sum_j \delta^g_j = 0$, and define:

$$r^{c\,*}_{i\omega} := -L^c_{i\omega} + \tfrac{1}{n+m}\,\ell^*(\omega) + \delta^c_i, \tag{111}$$

$$r^{g\,*}_{j\omega} := -L^g_{j\omega} + \tfrac{1}{n+m}\,\ell^*(\omega) + \delta^g_j. \tag{112}$$

Insurance market clearing (95) *holds by* (110)*:*

$$\sum_i r^{c\,*}_{i\omega} + \sum_j r^{g\,*}_{j\omega} = -\ell^*(\omega) + \ell^*(\omega) + 0 = 0.$$

Moreover, each agent's insured loss becomes:

$$L^c_{i\omega} + r^{c\,*}_{i\omega} = \tfrac{1}{n+m}\,\ell^*(\omega) + \delta^c_i, \tag{113}$$

which is an affine increasing function of the aggregate planner loss $\ell^*(\omega)$ with the same positive coefficient for all agents. Consequently, all agents share the same set of tail scenarios (those in which $\ell^*(\omega)$ exceeds its $(1-\beta)$-quantile), ensuring that the complementary slackness conditions on the CVaR constraints coincide across all agents and match the planner's (77)*.*

Proof of Walras' Law (WL). The day-ahead and real-time Walras' law conditions (104)–(105) follow from complementary slackness on the planner's balance constraints (60)–(61), exactly as in the risk-neutral case (Theorem 3):

$$\lambda^*\Big(\sum_j y^*_j - \sum_i x^*_i\Big) = 0 \implies p^*_0 \cdot z_0 = 0,$$

since $p^*_0 = \lambda^*/(1+\varepsilon)$, and similarly for each ω. The insurance Walras' law (106) holds by construction: the transfers (111)–(112) satisfy exact clearing (95), so the left-hand side of (106) is $q^*_\omega \cdot 0 = 0$. □

Remark 8 (Price interpretation). *The price formulas* (107) *admit a natural economic interpretation. The day-ahead price $p^*_0 = \lambda^*$ is the planner's marginal value of day-ahead energy. The real-time price $p^*_\omega = v^*_\omega/\tilde{\rho}^*_\omega$ is the marginal value of real-time energy in scenario ω, normalized by the CVaR-tilted weight rather than the raw probability; this reduces to v^*_ω/ρ_ω when $\varepsilon = 0$. The insurance price $q^*_\omega = \tilde{\rho}^*_\omega = (1-\varepsilon)\rho_\omega + \gamma^*_\omega$ is the standard* Arrow–Debreu state price*: the cost today of one unit of consumption tomorrow contingent on scenario ω. It decomposes into a fair-pricing component $(1-\varepsilon)\rho_\omega$ and a tail-risk premium γ^*_ω that is strictly positive only on scenarios in the CVaR tail. Because $\sum_\omega q^*_\omega = 1$, the insurance prices form a probability measure on Ω – the equilibrium worst-case (or risk-neutral pricing) measure under which the planner's allocation can be evaluated as a risk-neutral optimum.*

Remark 9 (Insurance transfers). *The construction* (111)–(112) *admits a clear interpretation: each agent's loss is pooled, and a pro-rata share of the aggregate loss $\ell^*(\omega)$ is redistributed. The constants δ^c_i, δ^g_j represent budget-neutral lump-sum transfers that can be chosen freely (subject to summing to zero) to satisfy any requirements without affecting the equilibrium conditions. This construction is explicit, in contrast to the pure existence arguments in [9] and [16].*

8.6 Welfare Theorems for Risk-Aware SCER

Lemma 5 (Common worst-case measure at any R-SCER). *Let* $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, \boldsymbol{Y}^*, r^{c*}, r^{g*}, p_0^*, \boldsymbol{p}^*, q^*)$ *be an R-SCER, and let* $\gamma^c_{i,\omega}, \gamma^g_{j,\omega}$ *be the multipliers on the agents' CVaR shortfall constraints* (97), (101). *Define*

$$\hat{Q}^*(\omega) \;:=\; q^*_\omega, \qquad \forall\, \omega \in \Omega.$$

Then $\hat{Q}^*$ *is a probability measure on* Ω *that lies in the risk set* $\boldsymbol{Q}_{RA}$ *of* ρ_{RA}, *and:*

(1) $\hat{Q}^*$ *is a worst-case measure for every agent's risk-adjusted utility: for each consumer* i *and each producer* j,

$$\rho_{RA}\big[-u_i(\cdot\,;\omega) - {r^c_{i\omega}}^*\big] = \mathop{\mathbb{E}}_{\hat{Q}^*}\big[-u_i(\cdot\,;\omega) - {r^c_{i\omega}}^*\big], \tag{114}$$

$$\rho_{RA}\big[-u_j(\cdot\,;\omega) - {r^g_{j\omega}}^*\big] = \mathop{\mathbb{E}}_{\hat{Q}^*}\big[-u_j(\cdot\,;\omega) - {r^g_{j\omega}}^*\big]. \tag{115}$$

(2) $\hat{Q}^*$ *is a worst-case measure for the aggregate planner loss:*

$$\rho_{RA}\big[\ell^*(\omega)\big] \;=\; \mathop{\mathbb{E}}_{\hat{Q}^*}\big[\ell^*(\omega)\big], \tag{116}$$

where $\ell^*(\omega) := -W2(\boldsymbol{x}^*, \boldsymbol{y}^*; \omega)$.

Proof. By insurance stationarity (108), the equilibrium first-order condition with respect to $r^c_{i\omega}$ gives $q^*_\omega = (1-\varepsilon)\rho_\omega + \gamma^c_{i,\omega}$, and analogously $q^*_\omega = (1-\varepsilon)\rho_\omega + \gamma^g_{j,\omega}$ for each producer. Hence every agent's CVaR multiplier is the same:

$$\gamma^c_{i,\omega} = \gamma^g_{j,\omega} = q^*_\omega - (1-\varepsilon)\rho_\omega =: \gamma^\dagger_\omega, \qquad \forall\, i, j, \omega.$$

The agents' threshold stationarity gives $\sum_\omega \gamma^c_{i,\omega} = \varepsilon$, so $\sum_\omega \gamma^\dagger_\omega = \varepsilon$, and consequently

$$\sum_\omega \hat{Q}^*(\omega) \;=\; \sum_\omega q^*_\omega \;=\; \sum_\omega \big[(1-\varepsilon)\rho_\omega + \gamma^\dagger_\omega\big] \;=\; (1-\varepsilon) + \varepsilon \;=\; 1,$$

so $\hat{Q}^*$ is a probability measure. The agents' slack stationarity gives $\gamma^\dagger_\omega \le \varepsilon\rho_\omega/(1-\beta)$. Defining $Q^*(\omega) := \gamma^\dagger_\omega/\varepsilon$, we have $\sum_\omega Q^*(\omega) = 1$ and $Q^*(\omega) \le \rho_\omega/(1-\beta)$, so $Q^* \in \boldsymbol{Q}_\beta$ and $\hat{Q}^* = (1-\varepsilon)P + \varepsilon Q^* \in \boldsymbol{Q}_{\mathrm{RA}}$.

To establish (114), observe that the agent's CVaR shortfall multipliers $\gamma^c_{i,\omega} = \gamma^\dagger_\omega = \varepsilon\, Q^*(\omega)$. This identifies Q^* as a worst-case measure in the dual representation for the CVaR applied to consumer i's insured loss $-u_i - {r^c_{i\omega}}^*$ (Lemma 4). Combining with the trivial identification of P as the worst-case measure for $\mathbb{E}[\cdot]$, Lemma 2 yields $\hat{Q}^*$ as a worst-case measure for ρ_{RA} applied to the agent's insured loss. The producer case (115) is identical. For the planner case (116), the same argument applies with the planner's CVaR multipliers γ^*_ω in place of $\gamma^\dagger_\omega$; that these coincide ($\gamma^*_\omega = \gamma^\dagger_\omega$) is precisely what the price identification $q^*_\omega = \tilde{\rho}^*_\omega$ delivers, and is verified directly through the planner's KKT in the proof of Theorem 7 below. □

The lemma states that at equilibrium, all market participants and the planner share a common worst-case measure $\hat{Q}^*$, and that this measure is given directly by the equilibrium insurance prices: $\hat{Q}^*(\omega) = q^*_\omega$. The Arrow–Debreu prices for state-contingent claims are not just any pricing system but are specifically the worst-case probability measure under which the entire risk-aware market behaves as risk-neutral. Under this common measure, every coherent risk measure in sight collapses to a linear expectation, restoring the additivity that makes the risk-neutral aggregation argument work.

Having established the existence of R-SCER via Theorem 6, we now state the welfare theorems for the risk-aware setting.

Theorem 7 (First Welfare Theorem for R-SCER). *If* $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, Y^*, r^{c*}, r^{g*}, p_0^*, \boldsymbol{p}^*, q^*)$ *is an R-SCER, then* $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, Y^*)$ *is a risk-adjusted welfare-maximizing allocation, that is, it solves* (66).

Proof. The argument parallels the proof of Theorem 4 for the risk-neutral case, with the dual representation of ρ_{RA} playing the role that linearity of expectation played in the risk-neutral setting.

Since the market outcome is an R-SCER, condition **(UM)** gives that each consumer i is utility-maximizing at $(x_i^*, \boldsymbol{X}_i^*, r_i^{c*})$ and each producer j is utility-maximizing at (y_j^*, Y_j^*, r_j^{g*}), given prices $(p_0^*, \boldsymbol{p}^*, q^*)$. Let $\mathcal{A}$ denote the program obtained by summing the agents' risk-adjusted utility-maximization objectives and aggregating their individual constraints. Since each agent's problem is solved independently at fixed prices, the equilibrium allocation $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, Y^*, r^{c*}, r^{g*})$ is an optimal solution to $\mathcal{A}$.

We show that $\mathcal{A}$ and the planner's problem (66) have the same optimal allocations, by verifying that they share the same objective and feasible set.

Objective. Each consumer i's objective at the equilibrium can be rewritten using ρ_{RA}:

$$(1-\varepsilon)\sum_{\omega}\rho_\omega\,(u_i + r_{i\omega}^c) - \sum_{\omega} q_\omega^* r_{i\omega}^c - \varepsilon\,\mathrm{CVaR}_\beta[-u_i - r_{i\omega}^c] \;=\; -\rho_{\mathrm{RA}}\big[-u_i - r_{i\omega}^c\big] - \sum_{\omega} q_\omega^* r_{i\omega}^c.$$

By Lemma 5, the worst-case measure $\hat{Q}^*$ collapses the risk measure to a linear expectation: $\rho_{\mathrm{RA}}[-u_i - r_{i\omega}^c] = \mathbb{E}_{\hat{Q}^*}[-u_i - r_{i\omega}^c]$. Therefore consumer i's objective equals

$$\mathop{\mathbb{E}}_{\hat{Q}^*}[u_i + r_{i\omega}^c] - \sum_{\omega} q_\omega^* r_{i\omega}^c \;=\; \mathop{\mathbb{E}}_{\hat{Q}^*}[u_i] + \sum_{\omega}\hat{Q}^*(\omega)\, r_{i\omega}^c - \sum_{\omega} q_\omega^* r_{i\omega}^c \;=\; \mathop{\mathbb{E}}_{\hat{Q}^*}[u_i],$$

where the final equality uses the identification $\hat{Q}^*(\omega) = q_\omega^*$ from Lemma 5: the expected payout of the insurance position under the worst-case measure exactly cancels the premium paid for it. The producer case is identical, with u_j in place of u_i.

Summing over all agents:

$$\sum_{i\in[n]}\mathop{\mathbb{E}}_{\hat{Q}^*}[u_i] + \sum_{j\in[m]}\mathop{\mathbb{E}}_{\hat{Q}^*}[u_j] \;=\; \mathop{\mathbb{E}}_{\hat{Q}^*}\Big[\sum_i u_i + \sum_j u_j\Big].$$

By the day-ahead and real-time balance constraints from **(F)**, the price-dependent terms in the quasi-linear utilities cancel when summed across consumers and producers, leaving $\sum_i u_i + \sum_j u_j = \mathrm{W2}(\boldsymbol{x}^*, \boldsymbol{y}^*; \omega) = -\ell^*(\omega)$. Hence the aggregated objective of $\mathcal{A}$ becomes

$$\mathop{\mathbb{E}}_{\hat{Q}^*}\big[-\ell^*(\omega)\big] \;=\; -\mathop{\mathbb{E}}_{\hat{Q}^*}\big[\ell^*(\omega)\big] \;=\; -\rho_{\mathrm{RA}}\big[\ell^*(\omega)\big],$$

where the last equality uses (116). Up to the sign convention (the planner minimizes $\rho_{\mathrm{RA}}[\ell^*]$ while $\mathcal{A}$ maximizes its negative), this is precisely the planner's objective in (66).

Constraints. The bound constraints (63), (64) and non-negativity (65) appear identically in each agent's individual problem and in the planner's problem. The market-clearing constraints (60), (61) and the insurance-clearing constraint (95) hold by condition **(F)**, and the planner's CVaR shortfall and threshold constraints (62) are satisfied with primal variables $t^* := \sum_k t_k^{c*}$ and $u_\omega{}^* := [\ell^*(\omega) - t^*]^+$.

Since $\mathcal{A}$ and the planner's problem share the same objective and feasible set, any optimal solution to $\mathcal{A}$ is also optimal for the planner. Therefore $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, Y^*)$ solves (66). □

Theorem 8 (Second Welfare Theorem for R-SCER). *If* $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, Y^*)$ *is a feasible welfare-maximizing allocation for the risk-aware planner problem* (66)*, then there exist prices* $(p_0^*, \boldsymbol{p}^*, q^*)$ *and insurance transfers* (r^{c*}, r^{g*}) *such that the market outcome* $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, Y^*, r^{c*}, r^{g*}, p_0^*, \boldsymbol{p}^*, q^*)$ *is an R-SCER.*

Proof. This is a direct consequence of Theorem 6. Equilibrium prices are constructed from the dual multipliers of the risk-aware planner problem via the identification (107), and the insurance transfers are constructed via (111)–(112). Theorem 6 establishes that all three R-SCER conditions are satisfied. □

Remark 10 (Risk-neutral and pure-CVaR limits). *When $\varepsilon = 0$, the CVaR penalty vanishes: $\gamma^*_\omega = 0$ for all ω, $\tilde{\rho}^*_\omega = \rho_\omega$, and $q^*_\omega = \rho_\omega$. The price identifications reduce to $p^*_0 = \lambda^*$ and $p^*_\omega = \nu^*_\omega/\rho_\omega$, recovering the risk-neutral formulas* (43) *for energy. The insurance market is priced at $q^*_\omega = \rho_\omega$. So, the premium $\sum_\omega q^*_\omega r^c_{i\omega}$ exactly equals the expected payout $\sum_\omega \rho_\omega r^c_{i\omega}$. Hence, in this case, the agents are indifferent, and the insurance market is redundant. The R-SCER reduces to the SCER of Definition 2 on the physical allocation, and the risk-neutral welfare theorems (Theorems 4–5) are special cases of Theorems 7–8.*

At the opposite extreme $\varepsilon = 1$, the planner optimizes pure CVaR. The expected-utility component drops from the agent's objective, and $q^*_\omega = \gamma^*_\omega$ since $(1-\varepsilon)\rho_\omega = 0$. All probability mass in $\hat{Q}^*$ comes from the CVaR tail, with $\hat{Q}^*$ supported on the worst $(1-\beta)$-fraction of scenarios. The Arrow–Debreu prices then concentrate entirely on tail scenarios, and trades in non-tail scenarios are free.

8.7 Existence of R-SCER under heterogeneous risk-preferences

[Ben: *Here is some more content on this topic, which I think is really important. Going to heterogeneous preference is what makes this model go from unrealistic to realistic. You should be explicit that there will be a problem in practice without resolving this: the risk aversion of the central planner may not be supported by the available risk structure of the individual players. So to get a socially desirable outcome, it may be necessary to add risk aversion into the system (i.e. the supply may be mostly risk neutral, the demand somewhat risk averse – but insufficiently so to match what the central planner wants. You may recall that I suggested that we would want to consider adding risk-adjustment agents to help deal with this problem in one of our meetings. That is, it may be necessary to add extra risk aversion into the market, through explicit "risk-aversion insurance-demanding" agent(s). They are the opposite of insurance because instead of accepting payment to handle catastrophe, they expect payment in a catastrophe in return for up front payment – they are in effect a big demander for insurance within the market (to make up for bidders under-insuring themselves relative to what the market maker wants). Such agents may require levy/subsidy, or maybe there is enough arbitrage opportunity for finance folks to want to fill this role – its not clear to me without a more thorough model of this works. I think the more signposting you can provide here to help the reader see the issues here, and what can be done to handle them in a realistic way will really help, and when doing this, being as realistic in assumption about real agents in real electricity markets as possible. How can the level of risk aversion of the bidders and the center match? It won't... If you can add transition material that places the Arrow-Debreu content of the prior section as orthogonal, in effect, to this topic that would also help. You may want "insurance transfers" via the prior section... but you still need the stuff here handled properly for this to fly in practice... As part of this you probably want to acknowledge the market participants may not support what the center wants (which I believe has the consequence of under-sized allocations at high prices – which isn't what you want in practice – so the addition of (possibly subsidized) pure finance agents can help fix this... You may want the full discussion of full finance agents to get its own discussion section after the math...*]

The results of Sections 8.5.1–8.6 assume that all agents share common risk parameters (ε, β). In practice, market participants may differ in their attitudes toward tail risk. A renewable generator exposed to weather uncertainty may adopt a different risk posture than a dispatchable thermal plant. [Ben: *Also buyer/seller will differ, and both may differ from the center.*] In this section, we

characterize precisely which forms of heterogeneity are compatible with competitive equilibrium and which are not.

8.7.1 Heterogeneous β and common ϵ. We now show that heterogeneous confidence levels β_i and β_j can be supported with a competitive equilibrium, provided the social planner adopts the appropriate confidence level.

Suppose each consumer i uses confidence level $\beta_i \in [0, 1)$ and each producer j uses $\beta_j \in [0, 1)$, while all agents (and the planner) share a common $\varepsilon \in [0, 1]$. Define

$$\beta_{\min} := \min\Big\{ \min_{i\in[n]} \beta_i,\ \min_{j\in[m]} \beta_j \Big\}, \tag{117}$$

and partition the set of agents into

$$S_{\min} := \{k : \beta_k = \beta_{\min}\}, \qquad S_{>} := \{k : \beta_k > \beta_{\min}\}.$$

Insurance stationarity (108) continues to give $q_\omega = (1-\varepsilon)\rho_\omega + \gamma^c_{k,\omega}$, so the agent's CVaR multiplier is $\gamma^c_{k,\omega} = q_\omega - (1-\varepsilon)\rho_\omega$. Summing over ω and using the agent's threshold stationarity $\sum_\omega \gamma^c_{k,\omega} = \varepsilon$ gives $\sum_\omega q_\omega = 1$ – the standard probability-measure constraint, common to every agent regardless of β_k. The slack stationarity, on the other hand, depends on β_k:

$$\gamma^c_{k,\omega} + \eta^c_{k,\omega} = \frac{\varepsilon\,\rho_\omega}{1-\beta_k}, \qquad \eta^c_{k,\omega} \geq 0, \tag{118}$$

giving the upper bound $\gamma^c_{k,\omega} \leq \varepsilon\,\rho_\omega/(1-\beta_k)$, equivalently

$$q_\omega \leq (1-\varepsilon)\,\rho_\omega + \frac{\varepsilon\,\rho_\omega}{1-\beta_k}. \tag{119}$$

The right-hand side is decreasing in β_k, so the tightest constraint is imposed by the agent with smallest β. The binding constraint reads

$$q_\omega \leq (1-\varepsilon)\,\rho_\omega + \frac{\varepsilon\,\rho_\omega}{1-\beta_{\min}}, \qquad \forall\,\omega \in \Omega, \tag{120}$$

which is precisely the constraint that arises from a planner using confidence level $\beta_{\min}$.

A critical structural consequence of (118) is that for any agent $k \in S_{>}$ (with $\beta_k > \beta_{\min}$), the slack dual ${\eta^c}_{k,\omega} > 0$ for every scenario — including tail scenarios — because $\gamma^c_{k,\omega} \leq \varepsilon\rho_\omega/(1-\beta_{\min}) < \varepsilon\rho_\omega/(1-\beta_k)$. Complementary slackness then forces ${\zeta^c_{k,\omega}}^* = 0$ for all ω. When the equilibrium tail multiplier $\gamma^c_{k,\omega} > 0$ as well, the CVaR shortfall constraint must bind with zero slack:

$$L^c_{k\omega} - r^*_{k\omega} = t^{c*}_k, \qquad \forall\, k \in S_{>},\ \forall\,\omega \in \Omega \text{ with } \gamma^c_{k,\omega} > 0. \tag{121}$$

Agents in $S_{>}$ require a *constant* insured loss across the entire tail. This rules out the equal-share comonotone construction (111)–(112), in which insured losses vary with $\ell^*(\omega)$, and necessitates a different transfer construction.

Theorem 9 (Existence under heterogeneous β). *Given common $\varepsilon \in [0, 1]$ and heterogeneous confidence levels $\{\beta_i\}_{i\in[n]}$, $\{\beta_j\}_{j\in[m]}$, let $(\boldsymbol{x}^*, \boldsymbol{y}^*, \boldsymbol{X}^*, \boldsymbol{Y}^*, t^*, u^*)$ solve the risk-aware planner problem (66) with confidence level $\beta_{\min}$ as defined in (117). Let $(\lambda^*, \nu^*, \gamma^*, \eta^*)$ be the associated optimal dual variables. Define prices via (107): $p^*_0 = \lambda^*$, $p^*_\omega = \nu^*_\omega/\tilde{\rho}^*_\omega$, $q^*_\omega = \tilde{\rho}^*_\omega = (1-\varepsilon)\rho_\omega + \gamma^*_\omega$.*

Then there exist insurance transfers ${r^c_{i\omega}}^*$ and ${r^g_{j\omega}}^*$ such that the market outcome is an R-SCER.

Proof. The price identification $q^*_\omega = \tilde{\rho}^*_\omega$ aligns each agent's tilted weight with the planner's, and β_k enters only through the slack dual $\eta^c_{k,\omega}$, which does not couple to physical variables. Dual feasibility $\eta^c_{k,\omega} \geq 0$ for each agent k follows from $\gamma^*_\omega \leq \varepsilon\rho_\omega/(1-\beta_{\min}) \leq \varepsilon\rho_\omega/(1-\beta_k)$.

Construction of insurance transfers. We construct one possible transfer that respect the structural requirement (121). Define each agent's per-scenario loss $L_{k\omega}$ at the equilibrium prices and the aggregate loss $\ell^*(\omega)$ as before, and set $t^* := \sum_k t_k^{c*}$. For each agent k, define:

$$r_{k\omega}^* = L_{k\omega} - t_k^{c*} - w_k\big(\ell^*(\omega) - t^*\big), \tag{122}$$

where the *risk-bearing weights* $w_k \geq 0$ satisfy $\sum_k w_k = 1$ and are chosen as:

$$w_k = \begin{cases} 0, & \text{if } k \in S_>, \\ 1/|S_{\min}|, & \text{if } k \in S_{\min}. \end{cases} \tag{123}$$

Market clearing. We verify (95):

$$\sum_k r_{k\omega}^* = \sum_k L_{k\omega} - \sum_k t_k^{c*} - \Big(\sum_k w_k\Big)\big(\ell^*(\omega) - t^*\big) = \ell^*(\omega) - t^* - \big(\ell^*(\omega) - t^*\big) = 0.$$

Insured losses. Each agent's insured loss equals

$$L_{k\omega} - r_{k\omega}^* = t_k^{c*} + w_k\big(\ell^*(\omega) - t^*\big). \tag{124}$$

For agents $k \in S_>$: $w_k = 0$, so the insured loss equals the constant t_k^{c*}, satisfying (121). For agents $k \in S_{\min}$: the insured loss is an affine increasing function of $\ell^*(\omega)$ with positive coefficient, so these agents share a common tail set with the planner.

CVaR complementary slackness. For agents $k \in S_>$: $\eta_{k,\omega}^c > 0$ for all ω, hence ${\zeta_{k,\omega}^c}^* = 0$, and the shortfall constraint reads $0 = L_{k\omega} - r_{k\omega}^* - t_k^{c*} = 0$, holding with equality. Both $\gamma_{k,\omega}^c \cdot 0 = 0$ and $\eta_{k,\omega}^c \cdot 0 = 0$ are satisfied. For agents $k \in S_{\min}$: the argument is identical to the homogeneous case (Theorem 6), since these agents share the planner's confidence level $\beta_{\min}$. □

[Hrithik: *Leaving this as a remark since my proof is still not verified.*]

Remark 11 (Heterogeneous ε and the general case). *The analysis extends to heterogeneity in both parameters. Insurance stationarity gives $\gamma_{k,\omega}^c = q_\omega - (1-\varepsilon_k)\rho_\omega$, so each agent's tilted weight equals q_ω regardless of ε_k, and the binding constraints on q are $q_\omega \geq (1-\varepsilon_{\min})\rho_\omega$ and $q_\omega \leq (1+\phi_{\min})\rho_\omega$, where $\phi_k := \varepsilon_k\beta_k/(1-\beta_k)$ and $\phi_{\min} := \min_k \phi_k$. These are exactly the constraints of a planner using $\varepsilon^P = \varepsilon_{\min}$ and $\beta^P = \phi_{\min}/(\varepsilon_{\min} + \phi_{\min})$; the mean–CVaR family is closed under intersection of risk sets, so the planner never leaves the family. A full treatment of the jointly heterogeneous case is deferred to the extended version of this paper.*

9 Experiments and Results

9.1 Synthetic Experiments

We first evaluate the risk-aware formulation in a controlled synthetic market with ten generators and one inelastic demand. Six generators are modeled as low-cost unreliable resources, representing renewable supply, while four generators are modeled as higher-cost reliable resources. Motivated by Huisman et al. [11], who show that variable renewables can generate heavy-tailed market outcomes, we model renewable uncertainty through common supply shock states: normal, mild shortfall, and severe shortfall. These shocks are applied across unreliable generators, inducing correlated low-renewable scenarios. The resulting market setup is summarized in Table 2 in Appendix A.

Figure 4(a) shows that this construction produces adverse low-welfare scenarios resulting from renewable shortfalls. When renewable output is normal, the cheap unreliable generators satisfy much of demand, so welfare is high. But when a common renewable shortfall occurs, multiple cheap generators lose capacity at the same time causing the market to rely on more expensive reliable generators or curtail demand at a high penalty. These low-welfare realizations are precisely

Table 1. Worst-5% tail welfare ($\times 10^7$) on IM3 GO WEST as wind generators (and hence uncertainty) increases with solar (3) and thermal (4) fixed

Wind Generators	Stochastic	CVaR	Deterministic
5	2.5836 ± 0.0064	**2.5874** ± 0.0064	2.5804 ± 0.0056
10	2.6183 ± 0.0234	**2.6270** ± 0.0246	2.6156 ± 0.0248
15	2.6163 ± 0.0217	**2.6257** ± 0.0224	2.6138 ± 0.0227
20	2.6488 ± 0.0206	**2.6590** ± 0.0210	2.6462 ± 0.0218

the outcomes that a risk-neutral objective may underweight but a CVaR objective is designed to internalize. We therefore compare deterministic clearing, risk-neutral stochastic clearing, and the CVaR-augmented formulation. The risk-neutral stochastic formulation achieves the highest expected welfare, reflecting its objective function. In contrast, the CVaR-augmented formulation improves mean welfare in the worst 5% of scenarios, indicating that risk aversion trades off some average efficiency for better protection against adverse tail outcomes, as shown in Figure 4(b,c).

[Ben: *Suggest noting here that you don't give up much in the expected case to get the big lift in the left-tail.*]

9.2 Empirical evaluation on IM3 GO WEST

We evaluate the same clearing rules on scenarios constructed from the IM3 GO WEST dataset [2], which provides time-indexed renewable generation and load profiles for a Western Interconnection test system. We use solar and wind profiles as unreliable generators and thermal units as reliable generators. We scale wind from 5 to 20 units, holding solar (3) and thermal (4) fixed, and report worst-5% tail welfare over 500 replications.

Table 1 shows that the CVaR-augmented formulation achieves the highest tail welfare across all tested configurations. The deterministic benchmark performs worst because its day-ahead schedule is optimized for a single forecasted system state, whereas the stochastic and CVaR formulations choose day-ahead commitments while anticipating real-time recourse across all scenarios. The gap between CVaR and risk-neutral stochastic clearing also increases in the larger renewable configuration, suggesting that the value of risk-aware clearing becomes more pronounced as exposure to renewable uncertainty grows.

To test robustness to risk preference, we additionally vary the CVaR weight $\epsilon_{\text{cvar}} \in \{0.1, 0.3, 0.5, 1\}$ at a fixed generator mix (5 wind, 3 solar, 4 thermal) and β=0.95. ~~Appendix B reports these results. Tail welfare gains over the risk-neutral benchmark grow monotonically with ϵ_{cvar}, at an increasing cost to expected welfare.~~The take away from this experiment is that the tail welfare gains over the risk-neutral benchmark grow monotonically with ϵ_{cvar}, at an increasing cost to expected welfare; see Appendix B for details.

The CVaR-augmented formulation also reallocates day-ahead commitment unevenly across renewable technologies: as wind's share grows, CVaR shifts commitment away from wind and toward solar, the lower-variance resource ~~as shown in Appendix C.~~; further details are provided in Appendix C.

10 Conclusions

In this work, we design a centralized mechanism for the electricity market that a risk-aware social planner can adopt, which delivers improved tail-welfare guarantees relative to its risk-neutral counterpart. To sustain a stochastic competitive equilibrium in a decentralized market populated by risk-aware agents, we show that it is necessary to introduce Arrow-Debreu securities, an "insurance

market" with trading instruments indexed to every scenario, thereby completing the market. Our analysis demonstrates that the day-ahead, real-time, and insurance prices can all be obtained as dual variables of the risk-aware social-welfare optimization problem solved by the planner, yielding a unified pricing framework that simultaneously reflects expected energy value, temporal flexibility across the two stages, and the cost of bearing risk in the tail of the scenario distribution. Further, we experimentally show how market-clearing using this risk-aware mechanism can yield better welfare in the tail events, thereby promising a more robust mechanism. Our experiment validates this using both synthetic experiments and data collected from a Western Interconnection test system.

A Synthetic Experiment Details

Table 2. Synthetic experiment setup.

Component	Specification
Number of generators	10
Unreliable generators	6
Reliable generators	4
Scenario probabilities	Dirichlet near-uniform, $\kappa = 1500$
Unreliable generator costs	Uniform on $[8, 14]$/MWh
Reliable generator costs	Uniform on $[35, 55]$/MWh
Renewable shock states	normal, mild shortfall, severe shortfall
Cost to curtail demand	\$1000/MWh

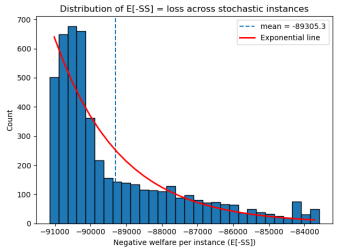


Fig. 1. Heavy-tailed welfare distribution

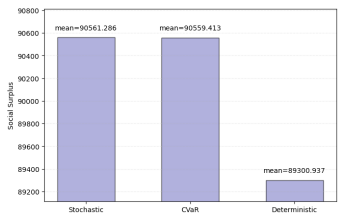


Fig. 2. Expected welfare

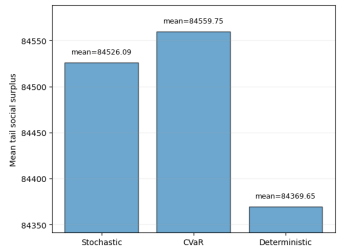


Fig. 3. Mean welfare in the worst 5% of scenarios

Fig. 4. Synthetic experiment diagnostics. Panel (a) shows that the renewable shortfall states induce a heavy-tailed welfare distribution. Panels (b) and (c) show that the CVaR-augmented formulation sacrifices some expected welfare relative to risk-neutral stochastic clearing in exchange for higher welfare in the worst 5% of scenarios, reducing exposure to catastrophic tail losses.

B Sensitivity to the Risk Preference Weight ϵ_{cvar}

Table 3. Expected social surplus and worst-5% tail welfare under CVaR clearing as ϵ_{cvar} varies, with 5 wind, 3 solar, and 4 thermal units fixed and $\beta = 0.95$. The Stochastic benchmark is shown for reference and does not depend on ϵ_{cvar}.

Method	E[Welfare] ($\times 10^7$)	Tail (5%) Welfare ($\times 10^7$)
Stochastic (Risk-neutral)	2.69164 ± 0.01418	2.58362 ± 0.00643
CVaR ($\epsilon_{\text{cvar}} = 0.1$)	2.69157 ± 0.01416	2.58487 ± 0.00611
CVaR ($\epsilon_{\text{cvar}} = 0.3$)	2.69090 ± 0.01405	2.58739 ± 0.00639
CVaR ($\epsilon_{\text{cvar}} = 0.5$)	2.68840 ± 0.01374	2.59120 ± 0.00583
CVaR ($\epsilon_{\text{cvar}} = 1$)	2.59380 ± 0.00548	2.59131 ± 0.00582

C CVaR-Induced Solar/Wind Substitution

The tail-welfare improvement reported in Section 1 is produced by a specific reallocation mechanism at the day-ahead stage. The CVaR-augmented clearing rule does not discount all renewable resources uniformly. Instead, it shifts commitment away from the higher-variance resource (wind) and, as wind's share of the portfolio grows, toward the more predictable resource (solar) [Hrithik: *In the IM3 dataset solar is predictable and constant since this is from the western part where mostly sunny.*]. Table 4 reports this effect in aggregate, averaged over the ten real-data instances in Section 1.

Table 4. Mean day-ahead reallocation under CVaR relative to risk-neutral stochastic clearing averaged over 10 IM3 GO WEST instances per configuration (solar = 3, thermal = 4 fixed).

Wind generators	Δ Solar (MW)	Δ Wind (MW)	Δ Total (MW)
5	−19.44	−86.78	−106.22
10	+18.43	−232.27	−213.84
15	+17.36	−266.23	−248.87

At the lowest wind penetration (5 wind generators), CVaR reduces day-ahead commitment for *both* solar and wind relative to the risk-neutral benchmark. Once wind capacity is large enough to represent a substantial share of the portfolio (10 and 15 wind generators), the sign on solar flips: the mean solar allocation increases under CVaR. This indicates that CVaR is not simply discounting renewable capacity uniformly, but reallocating within it according to relative risk.

Table 5 shows that CVaR clears the day-ahead market at a $100/MWh premium over the stochastic benchmark. The additional revenue does not fund more day-ahead delivery, since total day-ahead quantity falls by 329 MWh. This reduction is concentrated entirely in wind (−490 MWh), while every solar unit's commitment increases (+161 MWh combined, as shown per generator in Table 6), and thermal, the fully reliable resource, is unaffected in either direction.

Table 5. Day-ahead allocation by technology, Stochastic vs. CVaR clearing, single instance (10 wind generators, β=0.95, ϵ_{cvar}=0.1).

	Stochastic	CVaR	Δ (CVaR − Stoch)
DA price π ($/MWh)	1000.00	1100.00	+100.00
E[RT price] ($/MWh)	999.98	989.96	−10.02
DA solar (MWh)	58.51	219.84	+161.33
DA wind (MWh)	3761.20	3270.88	−490.32
DA thermal (MWh)	27023.72	27023.72	0.00
DA total (MWh)	30843.43	30514.44	−328.99

Table 6 also shows why this reallocation matters for tail exposure. Under Stochastic clearing, five wind units, including the three largest (wind_1, wind_7, wind_10), have negative expected RT−DA, meaning their day-ahead commitment exceeds expected real-time delivery. This produces a systematic shortfall concentrated in the highest-volume units. Under CVaR, the reduced day-ahead commitment is enough to flip all ten wind units to a positive RT−DA gap, eliminating this shortfall exposure at the cost of the day-ahead premium. This shift away from the higher-variance resource, and the resulting removal of shortfall exposure in the largest wind units, underlies the tail-welfare improvement reported in Table 1.

Table 6. Per-generator day-ahead commitment and expected real-time deviation, same instance as Table 5. E[RT] is identical across methods; RT−DA is negative (shortfall) for five wind units under Stochastic clearing but positive (surplus) for all ten under CVaR.

Generator	DA (Stoch)	DA (CVaR)	E[RT]	RT−DA (Stoch)	RT−DA (CVaR)
solar_1	21.26	94.82	139.31	+118.05	+44.49
solar_2	28.00	67.00	62.00	+34.00	−5.00
solar_3	9.25	58.02	63.97	+54.72	+5.95
wind_1	642.73	527.71	587.68	−55.06	+59.97
wind_2	102.00	81.00	141.06	+39.06	+60.06
wind_3	209.60	165.35	191.58	−18.01	+26.24
wind_4	5.64	5.17	7.74	+2.10	+2.57
wind_5	125.96	113.10	134.79	+8.83	+21.69
wind_6	214.53	193.16	237.25	+22.71	+44.09
wind_7	652.36	573.71	600.64	−51.72	+26.93
wind_8	165.62	149.65	152.49	−13.13	+2.83
wind_9	16.60	15.00	24.57	+7.97	+9.57
wind_10	1626.16	1447.02	1482.48	−143.68	+35.46

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